

Foundations of Quantum Mechanics Notes

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1 Hilbert Spaces

1.1 Motivation

As we've well known,

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{ipx/\hbar} \phi(p) = \sum_n c_n u_n(x),$$

where the wavefunction is respectively written in position, momentum, and energy basis. These different bases also share the same **normalizable / square integrable condition**:

$$\int_{-\infty}^{\infty} dx \psi^*(x) \psi(x) = \int_{-\infty}^{\infty} dp \phi^*(p) \phi(p) = \sum_n c_n^* c_n < \infty.$$

With proper normalization,

$$P = \psi^*(x) \psi(x), \quad \phi^*(p) \phi(p), \quad c_n^* c_n \quad (1)$$

is the probability (density).

All these being said, we realize that the Hilbert spaces $\{\psi(x)\}$, $\{\phi(p)\}$, $\{c_1, \dots, c_n\}$ are all equally good starting points for describing quantum mechanics (isomorphism). None of them is more fundamental than another.

1.2 Hilbert Spaces

The state of a quantum system is given by a vector in an abstract Hilbert space $\mathcal{H}$, and the different wavefunctions $\psi(x)$, $\phi(p)$, $(c_1, \dots, c_n)$ then correspond to expansion coefficients with respect to some basis of the complex vector space.

A Hilbert space $\mathcal{H}$ is a complex vector space with an inner product (Cauchy complete).

1.3 Complex Vector Space & Dual Space

1.3.1 Vector Space

- Closure: Additive / Multiplicative

$$u + v \in V, \lambda v \in V \text{ for } u, v \in V, \lambda \in \mathbb{F}.$$

- Associativity: Additive / Multiplicative

$$(u + v) + w = u + (v + w), (\lambda\mu)v = \lambda(\mu v) \text{ for } u, v, w \in V, \lambda, \mu \in \mathbb{F}.$$

- Identity: Additive / Multiplicative

$$0 + v = v, 1v = v.$$

- Inverse: Additive

$$\exists w \in V \text{ such that } v + w = 0 \text{ for } v \in V.$$

- Distributivity:

$$\lambda(u + v) = \lambda u + \lambda v, (\lambda + \mu)v = \lambda v + \mu v \text{ for } u, v \in V, \lambda, \mu \in \mathbb{F}.$$

1.3.2 Vector Subspace

A set W is a vector subspace if $W \subset V$ and W is a vector space. In order to check that W is really a subspace, we need to check

- Closure: Additive / Multiplicative;
- Additive Identity.

1.3.3 Dual Vector Space

The dual vector space V^* is the set of linear maps $f : V \rightarrow \mathbb{C}$ for an arbitrary vector space V :

$$\mathbf{v} \in V \mapsto f(\mathbf{v}) \in \mathbb{C}. \quad (2)$$

As a reminder, a linear map f satisfies

$$f(\alpha \mathbf{v} + \beta \mathbf{w}) = \alpha f(\mathbf{v}) + \beta f(\mathbf{w}). \quad (3)$$

- Addition: $(f_1 + f_2)(\mathbf{v}) = f_1(\mathbf{v}) + f_2(\mathbf{v})$.
- Scalar Multiplication: $(\alpha f)(\mathbf{v}) = \alpha f(\mathbf{v})$.
- Identity: $0(\mathbf{v}) = 0$.

1.3.4 Inner Product

An inner product on a complex vector space V is a binary composition $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$ that satisfies

- Additivity in the 2nd slot:

$$\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle. \quad (4)$$

- Homogeneity in the 2nd slot:

$$\langle \mathbf{u}, \lambda \mathbf{v} \rangle = \lambda \langle \mathbf{u}, \mathbf{v} \rangle. \quad (5)$$

- Conjugate symmetry:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle^*. \quad (6)$$

- Positivity:

$$\langle \mathbf{u}, \mathbf{u} \rangle \geq 0. \quad (7)$$

- Definiteness:

$$\langle \mathbf{u}, \mathbf{u} \rangle = 0 \Leftrightarrow \mathbf{u} = 0. \quad (8)$$

The following properties appear as the corollary:

- Anti-linearity in the 1st slot:

$$\langle \lambda \mathbf{u} + \mu \mathbf{v}, \mathbf{w} \rangle = \lambda^* \langle \mathbf{u}, \mathbf{w} \rangle + \mu^* \langle \mathbf{v}, \mathbf{w} \rangle. \quad (9)$$

- Norm:

$$\|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2}. \quad (10)$$

- Orthogonality:

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0. \quad (11)$$

1.3.5 Dual Correspondence

The dual correspondence is the map $\dagger: V \rightarrow V^*$ such that $\mathbf{v} \in V$ is taken to $\mathbf{v}^\dagger \in V^*$ with $\mathbf{v}^\dagger(\mathbf{w}) = \langle \mathbf{v}, \mathbf{w} \rangle$.

As we know $(\alpha \mathbf{v})^\dagger(\mathbf{w}) = \langle \alpha \mathbf{v}, \mathbf{w} \rangle = \alpha^* \langle \mathbf{v}, \mathbf{w} \rangle$,

$$(\alpha \mathbf{v})^\dagger = \alpha^* \mathbf{v}^\dagger. \quad (12)$$

1.4 Bra-ket Notation

Every bra $\langle \psi| \in \mathcal{H}^*$ is the $\dagger$ of a unique ket $|\psi\rangle \in \mathcal{H}$, and vice versa. Therefore,

$$\langle |\psi\rangle, |\phi\rangle \rangle = (|\psi\rangle)^\dagger |\phi\rangle \equiv \langle \psi|\phi\rangle. \quad (13)$$

1.5 Schwarz Inequality

$$|\langle \psi|\phi\rangle|^2 \leq \langle \psi|\psi\rangle \langle \phi|\phi\rangle \quad \text{with equality iff} \quad |\psi\rangle \propto |\phi\rangle. \quad (14)$$

Proof. Let $|\gamma\rangle = |\psi\rangle + \lambda |\phi\rangle$, $\lambda \in \mathbb{C}$. By positivity,

$$\begin{aligned}\langle\gamma|\gamma\rangle &= (\langle\psi| + \lambda^* \langle\phi|) (|\psi\rangle + \lambda |\phi\rangle) \\ &= \langle\psi|\psi\rangle + \lambda \langle\psi|\phi\rangle + \lambda^* \langle\phi|\psi\rangle + \lambda^* \lambda \langle\phi|\phi\rangle \geq 0.\end{aligned}$$

Let $\lambda = -\frac{\langle\phi|\psi\rangle}{\langle\phi|\phi\rangle}$. Thus,

$$\begin{aligned}\langle\gamma|\gamma\rangle &= \langle\psi|\psi\rangle + \lambda \langle\psi|\phi\rangle \\ &= \langle\psi|\psi\rangle - \frac{\langle\phi|\psi\rangle}{\langle\phi|\phi\rangle} \langle\psi|\phi\rangle \geq 0.\end{aligned}$$

By rearrangement,

$$|\langle\psi|\phi\rangle|^2 \leq \langle\psi|\psi\rangle \langle\phi|\phi\rangle. \quad (15)$$

□

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2 Linear Operators on Hilbert Space

2.1 Linear Operators

A linear operator is defined as a map $A : \mathcal{H} \rightarrow \mathcal{H}$, $|\psi\rangle \mapsto A|\psi\rangle$.

- Linearity:

$$A(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha A|\psi\rangle + \beta A|\phi\rangle. \quad (16)$$

- Addition and scalar multiplication of linear operators:

$$(A + B)|\psi\rangle = A|\psi\rangle + B|\psi\rangle, \quad (17)$$

$$(\alpha A)|\psi\rangle = \alpha A|\psi\rangle. \quad (18)$$

- Product of linear operators:

$$(AB)|\psi\rangle = A(B|\psi\rangle). \quad (19)$$

- Commutators: The product of linear operators is associative but not commutative.

$$\begin{cases} \text{Commutator:} & [A, B] = AB - BA \\ \text{Anticommutator:} & \{A, B\} = AB + BA \end{cases} \quad (20)$$

- Identity operator and inverse operator:

$$\mathbb{1}|\psi\rangle = |\psi\rangle \quad \Rightarrow \quad AA^{-1} = A^{-1}A = \mathbb{1} \quad (21)$$

$$(AB)^{-1} = B^{-1}A^{-1} \quad (22)$$

Proof. $(AB)^{-1}AB = \mathbb{1} \Rightarrow (AB)^{-1}A = B^{-1}$. Right multiply by A^{-1} , we see $(AB)^{-1} = B^{-1}A^{-1}$. $\square$

- Unique Zero Operator:

$$C|\psi\rangle = 0, \forall |\psi\rangle \in \mathcal{H} \Rightarrow C = 0.$$

The corollary deals with thinking of $C = A - B$. Therefore, $A|\psi\rangle = B|\psi\rangle, \forall |\psi\rangle \in \mathcal{H} \Rightarrow A = B$.

- Alternative definition:

$A : \mathcal{H}^* \rightarrow \mathcal{H}^*$, $\langle\psi| \mapsto \langle\psi|A$. This gives us two equivalent ways of interpreting the following expression:

$$\langle\psi|A|\phi\rangle = (\langle\psi|A)|\phi\rangle = \langle\psi|(A|\phi\rangle). \quad (23)$$

2.2 Hermitian Conjugation

In definition, the Hermitian conjugate $A^\dagger$ of a linear operator A is

$$A^\dagger |\psi\rangle = (\langle\psi| A)^\dagger \Rightarrow (A^\dagger |\psi\rangle)^\dagger = \langle\psi| A. \quad (24)$$

Noting that

$$|\phi\rangle^\dagger |\psi\rangle = \langle\phi|\psi\rangle = \langle\psi|\phi\rangle^* = \left(|\psi\rangle^\dagger |\phi\rangle\right)^*, \quad (25)$$

$$\langle\phi|A^\dagger|\psi\rangle = |\phi\rangle^\dagger (A^\dagger |\psi\rangle) = \left[(A^\dagger |\psi\rangle)^\dagger |\phi\rangle\right]^* = \langle\psi|A|\phi\rangle^*.$$

We get a key alternative definition of the Hermitian conjugate:

$$\langle\phi|A^\dagger|\psi\rangle = \langle\psi|A|\phi\rangle^*. \quad (26)$$

The two pieces of corollary follow.

$$(A^\dagger)^\dagger = A^{\dagger\dagger} = A \quad (27)$$

Proof. By definition, $\langle\phi|A^{\dagger\dagger}|\psi\rangle = \langle\psi|A^\dagger|\phi\rangle^* = \langle\phi|A|\psi\rangle^{**} = \langle\phi|A|\psi\rangle$. $\square$

$$(AB)^\dagger = B^\dagger A^\dagger \quad (28)$$

Proof. By definition, $\langle\phi|(AB)^\dagger|\psi\rangle = \langle\psi|AB|\phi\rangle^* = \left[(A^\dagger |\psi\rangle)^\dagger B |\phi\rangle\right]^* = (B |\phi\rangle)^\dagger A^\dagger |\psi\rangle = \langle\phi|B^\dagger A^\dagger|\psi\rangle$. $\square$

2.3 Outer Product

For kets $|\psi\rangle$ and $|\phi\rangle \in \mathcal{H}$, we may construct an operator $|\phi\rangle\langle\psi| : \mathcal{H} \rightarrow \mathcal{H}$: $(|\phi\rangle\langle\psi|) |\gamma\rangle = |\phi\rangle \langle\psi|\gamma\rangle$.

- Linearity:

$$(|\phi\rangle\langle\psi|) (\lambda |\alpha\rangle + \mu |\beta\rangle) = |\phi\rangle (\lambda \langle\psi|\alpha\rangle + \mu \langle\psi|\beta\rangle) = \lambda (|\phi\rangle\langle\psi|) |\alpha\rangle + \mu (|\phi\rangle\langle\psi|) |\beta\rangle.$$

- Adjoint:

$$\langle\alpha| (|\phi\rangle\langle\psi|)^\dagger |\beta\rangle = (\langle\beta|\phi\rangle \langle\psi|\alpha\rangle)^* = \langle\phi|\beta\rangle \langle\alpha|\psi\rangle = \langle\alpha|\psi\rangle \langle\phi|\beta\rangle \Rightarrow (|\phi\rangle\langle\psi|)^\dagger = |\psi\rangle\langle\phi|.$$

To take the adjoint of a string, simply take the adjoint of each factor and reverse the order.

2.4 Bases

2.4.1 Definition

A **basis** for a finite dimensional Hilbert space is a set of vectors such that any $|\psi\rangle \in \mathcal{H}$ can be expanded uniquely.

In our sense, uniquely means that the basis vectors $|n\rangle$ all are linearly independent:

$$|\psi\rangle = \sum_{n=1}^N c_n |n\rangle. \quad (29)$$

2.4.2 Orthonormality

$$\langle n|m\rangle = \delta_{nm} \quad (30)$$

2.4.3 Resolution of the Identity / Completeness Relation

In an orthonormal basis, $|\psi\rangle = \sum_{n=1}^N c_n |n\rangle$. So,

$$\langle m|\psi\rangle = \sum_n c_n \langle m|n\rangle = \sum_n c_n \delta_{mn} = c_m.$$

This gives us

$$|\psi\rangle = \sum_n \langle n|\psi\rangle |n\rangle = \left(\sum_n |n\rangle \langle n| \right) |\psi\rangle.$$

Therefore,

$$\mathbb{1} = \sum_n |n\rangle \langle n|. \quad (31)$$

2.5 Hermitian and Unitary Operators

2.5.1 Hermitian Operators

A linear operator A is Hermitian if

$$A^\dagger = A \quad \Leftrightarrow \quad \langle \psi|A|\phi\rangle = \langle \phi|A|\psi\rangle^*, \quad \forall |\psi\rangle, |\phi\rangle \in \mathcal{H}. \quad (32)$$

2.5.2 Anti-Hermitian Operators

$$A^\dagger = -A, \quad \text{Anti-Hermitian} = i \cdot \text{Hermitian} \quad (33)$$

2.5.3 Unitary Operators

A linear operator U is unitary if

$$UU^\dagger = U^\dagger U = \mathbf{1} \quad \Leftrightarrow \quad U^\dagger = U^{-1}. \quad (34)$$

2.5.4 Corollary

$$\text{If } A^\dagger = A, B^\dagger = B, \text{ then } (AB)^\dagger = AB \quad \Leftrightarrow \quad [A, B] = 0. \quad (35)$$

Proof. To prove equivalence, we imply that the truth of either statement leads to the other. Firstly, if LHS is true,

$$\begin{aligned} [A, B] &= AB - BA \\ &= AB - B^\dagger A^\dagger \quad (A, B \text{ are Hermitian}) \\ &= AB - (AB)^\dagger = 0 \quad \text{if } (AB)^\dagger = AB. \end{aligned}$$

Then, if RHS is true,

$$(AB)^\dagger = B^\dagger A^\dagger = BA = AB \quad \text{if } [A, B] = 0.$$

□

$$U_1, U_2 \text{ unitary} \quad \Rightarrow \quad U_1 U_2 \text{ unitary}. \quad (36)$$

Proof. By unitarity, we know $U^\dagger = U^{-1}$. Thus,

$$(U_1 U_2)^\dagger = U_2^\dagger U_1^\dagger = U_2^{-1} U_1^{-1} = (U_1 U_2)^{-1}.$$

□

2.5.5 Theorems

- A Hermitian operator always has **real eigenvalues** and **orthogonal eigenvectors**.

Proof. Let $A|n\rangle = \lambda_n|n\rangle$. So,

$$\langle m|A|n\rangle = \lambda_n \langle m|n\rangle = \lambda_m^* \langle m|n\rangle \quad \Rightarrow \quad (\lambda_m^* - \lambda_n) \langle m|n\rangle = 0.$$

If $m = n$, positivity says that $\langle n|n\rangle \neq 0$ for $|n\rangle \neq 0$, so $\lambda_n^* = \lambda_n$ (real eigenvalues).

If $m \neq n$, we assume $\lambda_m \neq \lambda_n$ such that $\langle m|n \rangle = 0$ (orthogonal eigenvectors). $\square$

- Spectral Theorem: If $A^\dagger = A$, then there exists an orthonormal basis of $\mathcal{H}$, $\{|n\rangle\}$, that consists of the eigenvectors of A : $A|n\rangle = a_n|n\rangle$, $n \in \mathbb{R}$. Thus,

$$A = A|n\rangle\langle n| = a_n|n\rangle\langle n|. \quad (37)$$

Note that this orthonormal basis isn't unique. There may be degeneracies, and every basis can be multiplied by an arbitrary phase factor. With degeneracy, $\langle n, r_n|m, r_m \rangle = \delta_{nm}\delta_{r_n r_m}$, and so

$$\mathbb{1} = \sum_n \sum_{r_n} |n, r_n\rangle\langle n, r_n|. \quad (38)$$

Also, as $A|n, r_n\rangle = a_n|n, r_n\rangle$,

$$A = A \sum_n \sum_{r_n} |n, r_n\rangle\langle n, r_n| = \sum_n a_n \sum_{r_n} |n, r_n\rangle\langle n, r_n|. \quad (39)$$

- If $A_i^\dagger = A_i$, $[A_i, A_j] = 0$, then there exists an orthonormal basis of $\mathcal{H}$ that consists of the simultaneous eigenvectors of all A_i .
Let $A_i|n\rangle = \lambda_{n_i}|n\rangle$. So, a complete set of commuting operators is one such that knowing λ_{n_i} for all i fixes $|n\rangle$ completely (i.e. no degeneracy).
- If $A^\dagger = A$, $B^\dagger = B$, $[A, B] = 0$, then any non-degenerate eigenvector of A is an eigenvector of B .

Proof. Let $A|n\rangle = \lambda|n\rangle$, $|n\rangle \neq 0$. So, $[A, B]|n\rangle = 0$ gives $AB|n\rangle = BA|n\rangle = \lambda_n B|n\rangle$. From this we see $A(B|n\rangle) = \lambda_n(B|n\rangle)$. As A has non-degenerate eigenvectors, we have $B|n\rangle \propto |n\rangle$, $B|n\rangle = \beta|n\rangle$. $\square$

2.6 Projection Operators: Hermitian & Idempotent

2.6.1 Definition

A projection operator P satisfies $P = P^\dagger$ and $P^2 = P$.

2.6.2 Eigenvalues

Let $P|\psi\rangle = \lambda|\psi\rangle$. Thus,

$$(P^2 - P)|\psi\rangle = [\lambda(\lambda - 1)]|\psi\rangle = 0,$$

which gives $\lambda = 0, 1$.

2.6.3 Orthonormal Bases

$P_n = |n\rangle\langle n| \Rightarrow P_n P_m = \delta_{nm} P_m$. Also, $\mathbb{1} = \sum_n P_n$. With degeneracies,

$$A = \sum_n \lambda_n P_n \quad \text{where} \quad P_n = \sum_{r_n} |n, r_n\rangle\langle n, r_n|. \quad (40)$$

2.7 Functions of a Hermitian Operator

Consider A^2 , $A|n\rangle = \lambda_n|n\rangle$:

$$\begin{aligned} A^2 &= A\mathbb{1}A\mathbb{1} \\ &= \sum_{nm} A|n\rangle\langle n|A|m\rangle\langle m| \\ &= \sum_{nm} \lambda_n \lambda_m |n\rangle\langle n|m\rangle\langle m| \\ &= \sum_{nm} \lambda_n \lambda_m \delta_{nm} |n\rangle\langle m| \\ &= \sum_n \lambda_n^2 P_n. \end{aligned}$$

With A^2 , we may take a step forward and consider A^3 :

$$\begin{aligned} A^3 &= A^2 A \\ &= \sum_{nm} \lambda_n^2 P_n \lambda_m P_m \\ &= \sum_{nm} \lambda_n^2 \lambda_m \delta_{nm} P_n \\ &= \sum_n \lambda_n^3 P_n. \end{aligned}$$

In conclusion, by Taylor expansion,

$$F(A) = \sum_n F(\lambda_n) P_n. \quad (41)$$

2.8 Operators, Bras, Kets v.s. Matrices, Rows, Columns

$$\begin{aligned} A &= \mathbb{1}A\mathbb{1} \\ &= \sum_{nm} |n\rangle\langle n|A|m\rangle\langle m| \\ &\equiv \sum_{nm} A_{nm} |n\rangle\langle m| \end{aligned}$$

In this way, we relate the matrix elements to an operator. We may also relate bras and kets to rows and columns:

$$\begin{aligned} |\psi\rangle &= \mathbb{1} |\psi\rangle = \sum_n |n\rangle \langle n|\psi\rangle \equiv \sum_n \psi_n |n\rangle, \\ \langle\psi| &= \langle\psi| \mathbb{1} = \sum_n \langle\psi|n\rangle \langle n| \equiv \sum_n \psi_n^* \langle n|. \end{aligned}$$

Therefore,

$$\langle\psi|A|\phi\rangle = \sum_{nm} \langle\psi|n\rangle \langle n|A|m\rangle \langle m|\phi\rangle = \sum_{nm} \psi_n^* A_{nm} \phi_m,$$

which resembles matrix multiplication.

With Hermitian A , $\langle m|A|n\rangle = \langle n|A|m\rangle^*$, so $A_{mn} = A_{nm}^*$, $A = A^\dagger$ in the language of matrices as well.

2.9 Change of Basis

Define $U : \mathcal{H} \rightarrow \mathcal{H}$, $U|n\rangle = |n'\rangle$. Assume that $\{|n\rangle\}$ and $\{|n'\rangle\}$ are orthonormal.

2.9.1 Unitary U

We have $\langle m|U^\dagger U|n\rangle = \langle m'|n'\rangle = \delta_{m'n'} = \delta_{mn}$. Since m and n are arbitrary, we see $U^\dagger U = \mathbb{1}$. So,

$$U^\dagger U|n\rangle = U^\dagger|n'\rangle = |n\rangle.$$

This implies that

$$\langle m'|UU^\dagger|n'\rangle = \langle m|n\rangle = \delta_{mn} = \langle m'|n'\rangle.$$

Therefore, we've proved that $UU^\dagger = U^\dagger U = \mathbb{1}$, so $U^\dagger = U^{-1}$.

2.9.2 Change of Basis

Define $\psi_n = \langle n|\psi\rangle$, $\psi'_n = \langle n'|\psi\rangle$, $A_{mn} = \langle m|A|n\rangle$, $A'_{mn} = \langle m'|A|n'\rangle$. By plugging in resolution of identity,

$$\begin{aligned} \psi'_n &= \langle n'|\psi\rangle \\ &= \sum_m \langle n'|U^\dagger|m\rangle \langle m|\psi\rangle \\ &= \sum_m U_{nm}^\dagger \psi_m \\ &= (U^\dagger \psi)_n = (U^{-1} \psi)_n, \end{aligned}$$

and similarly,

$$\begin{aligned} A'_{mn} &= \langle m'|A|n'\rangle \\ &= \sum_{ij} \langle m|U^\dagger|i\rangle \langle i|A|j\rangle \langle j|U|n\rangle \\ &= \sum_{ij} U_{mi}^\dagger A_{ij} U_{jn} \\ &= (U^\dagger AU)_{mn} = (U^{-1}AU)_{mn}. \quad (\text{unitary transformation}) \end{aligned}$$

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3 Postulates of Quantum Mechanics

3.1 Postulates

- A quantum system is associated with a Hilbert space $\mathcal{H}$.
- Observables correspond to Hermitian operators, and possible results correspond to eigenvalues.
- **The Born rule:**

$$P(A = a_n) = \frac{\langle \psi | P_n | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\|P_n | \psi \rangle\|^2}{\| | \psi \rangle \|^2}. \quad (42)$$

With degeneracies,

$$A = \sum_n a_n P_n \quad \text{where} \quad P_n = \sum_{r_n} |n, r_n\rangle \langle n, r_n|. \quad (43)$$

- After a measurement with outcome a_n , the system collapses into the state $P_n | \psi \rangle$:

$$P_n | \psi \rangle = \sum_{r_n} |n, r_n\rangle \langle n, r_n | \psi \rangle. \quad (44)$$

- Time evolution: **in between measurements**

$$i\hbar \frac{\partial}{\partial t} | \psi, t \rangle = H | \psi, t \rangle \quad (45)$$

There are several important interpretations of these postulates.

- Rays:

A ray is the one-dimensional subspace in $\mathcal{H}$ defined by $\{\alpha | \psi \rangle\}$, $\alpha \in \mathbb{C}$ with $| \psi \rangle \neq 0$. **All kets in a ray are physically equivalent.**

The state of a quantum system is specified by a ray, but not an individual ket.

Even normalization doesn't fix an individual ket; given $| \psi \rangle_n = \frac{| \psi \rangle}{\langle \psi | \psi \rangle}$, $| \psi \rangle_n$ and $e^{i\theta} | \psi \rangle_n$ are physically equivalent.

- Quantum mechanics makes statistical predictions on an ensemble.
- Every measurement is a preparation of a new state. After a measurement, the state is in general not normalized.
- In preparation and measurement, the state evolves non-deterministically. However, in between it evolves deterministically via the Schrödinger equation.
- Commuting observables can be measured simultaneously.

3.2 Simultaneous Measurement

To measure a value a of the observable A , the probability is

$$P(A = a) = \frac{\langle \psi | P_a | \psi \rangle}{\langle \psi | \psi \rangle}.$$

After this measurement, the system collapses to $|\tilde{\psi}\rangle = P_a |\psi\rangle$. Then, we may calculate the conditional probability of then measuring $B = b$:

$$P(B = b | A = a) = \frac{\langle \tilde{\psi} | P_b | \tilde{\psi} \rangle}{\langle \tilde{\psi} | \tilde{\psi} \rangle} \cdot \frac{\langle \psi | P_a | \psi \rangle}{\langle \psi | \psi \rangle}.$$

However, as we know about the projection operator,

$$\langle \tilde{\psi} | \tilde{\psi} \rangle = \langle \psi | P_a^\dagger P_a | \psi \rangle = \langle \psi | P_a | \psi \rangle, \quad (46)$$

we know that

$$P(B = b | A = a) = \frac{\langle \psi | P_a P_b P_a | \psi \rangle}{\langle \psi | \psi \rangle}. \quad (47)$$

Typically, $P(B = b | A = a) \neq P(A = a | B = b)$ unless $[A, B] = 0$.

3.3 Lemma: Projection Operators

$$[A, B] = 0 \quad \Leftrightarrow \quad [P_{a_n}, P_{b_m}] = 0, \quad \forall n, m. \quad (48)$$

Proof. As we know $A = \sum_n a_n P_{a_n}$ and $B = \sum_m b_m P_{b_m}$ from the spectral theorem,

$$[A, B] = \left[\sum_n a_n P_{a_n}, \sum_m b_m P_{b_m} \right] = \sum_{nm} a_n b_m [P_{a_n}, P_{b_m}].$$

□

3.4 Expectation Value and Variance

3.4.1 Expectation Value

$$\langle A \rangle = \langle \psi | A | \psi \rangle \quad (49)$$

for a normalized state $|\psi\rangle$. If this state isn't normalized, the expectation value should be divided by its norm squared:

$$\langle A \rangle = \frac{\langle \psi | A | \psi \rangle}{\langle \psi | \psi \rangle}. \quad (50)$$

By the spectral theorem, $A = \sum_n a_n P_n$,

$$\langle A \rangle = \sum_n a_n \langle \psi | P_n | \psi \rangle = \sum_n a_n P(A = a_n). \quad (51)$$

3.4.2 Variance

$$\Delta A^2 = \langle \psi | (A - \langle A \rangle)^2 | \psi \rangle = \langle A^2 \rangle - \langle A \rangle^2 \quad (52)$$

for a normalized state $|\psi\rangle$.

3.5 Generalized Uncertainty Principle

$$\Delta A^2 \Delta B^2 \geq \frac{1}{4} \langle [A, B] \rangle^2 \quad (53)$$

Proof. Let $|\alpha\rangle = (A - \langle A \rangle) |\psi\rangle = A_1 |\psi\rangle$, $|\beta\rangle = (B - \langle B \rangle) |\psi\rangle = B_1 |\psi\rangle$. As A and B are observables, A_1 and B_1 are as well Hermitian. Now

$$A_1 B_1 = \frac{1}{2} (A_1 B_1 + B_1 A_1) + \frac{1}{2} (A_1 B_1 - B_1 A_1) = \frac{1}{2} ([A_1, B_1] + \{A_1, B_1\}),$$

and thus

$$\langle \alpha | \beta \rangle = \langle \psi | A_1 B_1 | \psi \rangle = \frac{1}{2} \langle [A_1, B_1] \rangle + \frac{1}{2} \langle \{A_1, B_1\} \rangle.$$

Since the commutator for A_1 and B_1

$$[A_1, B_1]^\dagger = (A_1 B_1)^\dagger - (B_1 A_1)^\dagger = B_1 A_1 - A_1 B_1 = -[A_1, B_1]$$

is anti-Hermitian, and the anticommutator is Hermitian, we know that $\langle \{A_1, B_1\} \rangle \in \mathbb{R}$ and $\langle [A_1, B_1] \rangle$ is purely imaginary:

$$\langle \alpha | \beta \rangle = x + iy$$

for $x, y \in \mathbb{R}$. Then, by Schwarz inequality,

$$\langle \alpha | \alpha \rangle \langle \beta | \beta \rangle \geq |\langle \alpha | \beta \rangle|^2 \geq y^2 = |iy|^2,$$

$$\Delta A^2 \Delta B^2 \geq \frac{1}{4} \langle [A_1, B_1] \rangle^2.$$

□

3.6 Saturated Uncertainty Principle

The equality holds in $\langle \alpha | \alpha \rangle \langle \beta | \beta \rangle \geq |\langle \alpha | \beta \rangle|^2$ only when $|\alpha\rangle = c |\beta\rangle$, $c \in \mathbb{C}$. Also, the real part, $\langle \{A_1, B_1\} \rangle$, should vanish. So,

$$\langle \{A_1, B_1\} \rangle = \langle \alpha | \beta \rangle + \langle \beta | \alpha \rangle = (c^* + c) \langle \beta | \beta \rangle = 0,$$

which implies that $c = i\mu$, $\mu \in \mathbb{R}$.

Therefore, $A_1 |\psi\rangle = -i\mu B_1 |\psi\rangle$, and by definition of A_1 and B_1 we get

$$(A + i\mu B) |\psi\rangle = (\langle A \rangle + i\mu \langle B \rangle) |\psi\rangle, \tag{54}$$

a characteristic for coherent states, which turn out to have maximal uncertainty.

4 Position Space Representation and Wavefunctions

4.1 The Position Operator

4.1.1 Definition

$$\hat{x} |x\rangle = x |x\rangle \quad (55)$$

The basis $\{|x\rangle : x \in \mathbb{R}\}$ is clearly uncountably infinite.

4.1.2 Orthonormality

$$\langle x|y\rangle = \delta(x-y) \quad \text{where} \quad \delta(x) = \delta(-x) \quad (56)$$

4.1.3 Resolution of the Identity

$$\mathbf{1} = \int dx |x\rangle\langle x| \quad (57)$$

4.1.4 Wavefunction

$$\psi(x) \equiv \langle x|\psi\rangle \quad (58)$$

4.1.5 Cauchy Completeness

$$\|\psi\|^2 = \int dx \langle \psi|x\rangle \langle x|\psi\rangle = \int dx \psi^*(x) \psi(x) < \infty \quad (59)$$

4.1.6 Projection Operator

$$P_I = \int_{x_0}^{x_1} dx |x\rangle\langle x| \quad \text{with interval} \quad I = [x_0, x_1] \quad (60)$$

We may check if this is really a projection operator by idempotency and Hermiticity:

$$P_I^\dagger = \int_{x_0}^{x_1} dx (|x\rangle\langle x|)^\dagger = P_I,$$

$$\begin{aligned}
P_I^2 &= \int_{x_0}^{x_1} dx \int_{x_0}^{x_1} dy |x\rangle \langle x|y\rangle \langle y| \\
&= \int_{x_0}^{x_1} dx \int_{x_0}^{x_1} dy \delta(x-y) |x\rangle \langle y| \\
&= \int_{x_0}^{x_1} dx |x\rangle \langle x| = P_I.
\end{aligned}$$

So, for normalized $|\psi\rangle$, the probability of x falling in this interval is

$$P(x \in I) = \langle \psi | P_I | \psi \rangle = \int_{x_0}^{x_1} dx \psi^*(x) \psi(x),$$

and the probability density is $\psi^*(x)\psi(x)$.

4.1.7 Inner Product

$$\langle \phi | \psi \rangle = \int dx \phi^*(x) \psi(x) \quad (61)$$

4.1.8 Expectation Value

$$\langle x \rangle = \langle \psi | \int dx \hat{x} |x\rangle \langle x| \psi \rangle = \int dx x \psi^*(x) \psi(x) \quad (62)$$

4.2 The Momentum Operator

4.2.1 Definition

The momentum operator is defined via

$$[\hat{x}, \hat{p}] = i\hbar. \quad (63)$$

This definition skips the fact that $\hat{p}$ corresponds to a physical observable, so we must return to check that $\hat{p}$ is Hermitian:

$$[\hat{x}, \hat{p}]^\dagger = (\hat{x}\hat{p})^\dagger - (\hat{p}\hat{x})^\dagger = \hat{p}^\dagger \hat{x} - \hat{x} \hat{p}^\dagger = -[\hat{x}, \hat{p}^\dagger] = -i\hbar = -[\hat{x}, \hat{p}].$$

The above argument proves that $\hat{p} = \hat{p}^\dagger$ up to a function of $\hat{x}$.

4.2.2 Fourier Transform

Now we are to derive the momentum wavefunction from scratch. Plugging the commutator $[\hat{x}, \hat{p}]$ in between the bracket $\langle x|y\rangle$, we get

$$\begin{aligned}\langle x|[\hat{x}, \hat{p}]|y\rangle &= i\hbar \langle x|y\rangle \\ \langle x|(\hat{x}\hat{p} - \hat{p}\hat{x})|y\rangle &= i\hbar\delta(x-y) \\ (x-y) \langle x|\hat{p}|y\rangle &= i\hbar\delta(x-y).\end{aligned}$$

Since we know from the properties of the delta function,

$$x\delta(x) = 0 \quad \Rightarrow \quad \delta(x) = -x\delta'(x). \quad (64)$$

Thus,

$$(x-y) \langle x|\hat{p}|y\rangle = -i\hbar(x-y)\delta'(x-y) + \alpha(x)(x-y)\delta(x-y),$$

where $\alpha(x) \in \mathbb{R}$ such that $\hat{p}$ is Hermitian. Now we can cancel $(x-y)$ on both sides:

$$\langle x|\hat{p}|y\rangle = -i\hbar\delta'(x-y) + \alpha(x)\delta(x-y). \quad (65)$$

For simplicity, we'll stick to $\alpha(x) = 0$, i.e. $\langle x|\hat{p}|y\rangle = -i\hbar\delta'(x-y)$. So,

$$\begin{aligned}(\hat{p}\psi)(x) &\equiv \langle x|\hat{p}|\psi\rangle \\ &= \langle x|\int dy \hat{p}|y\rangle \langle y|\psi\rangle \\ &= -i\hbar \int dy \delta'(x-y)\psi(y) \quad \left(\delta'(x-y) = \frac{d}{d(x-y)}\delta(x-y) = -\frac{d}{dy}\delta(x-y) \right) \\ &= i\hbar \left[\delta(x-y)\psi(y) \right]_{-\infty}^{\infty} - \int dy \frac{d}{dy}\psi(y)\delta(x-y) \\ &= -i\hbar \frac{d}{dx}\psi(x).\end{aligned} \quad (66)$$

Remember, $\hat{p} \neq -i\hbar \frac{d}{dx}$. Now let's meet the momentum space, which is very similar to the position space:

$$\hat{p}|p\rangle = p|p\rangle, \quad \mathbb{1} = \int dp |p\rangle\langle p|, \quad |\psi\rangle = \int dp |p\rangle \langle p|\psi\rangle = \int dp |p\rangle \psi(p). \quad (67)$$

Therefore, $\langle x|\hat{p}|p\rangle = -i\hbar \frac{d}{dx} \langle x|p\rangle = p \langle x|p\rangle$, a differential equation with solution $\langle x|p\rangle = C e^{ipx/\hbar}$. Now

$$\begin{aligned}\langle x|y\rangle &= \int dp \langle x|p\rangle \langle p|y\rangle \\ &= |C|^2 \int dp e^{ipx/\hbar} e^{-ipy/\hbar} \\ &= |C|^2 2\pi\hbar \cdot \frac{1}{2\pi} \int d(p/\hbar) e^{ip(x-y)/\hbar} \\ &= |C|^2 2\pi\hbar \delta(x-y) = \delta(x-y),\end{aligned}$$

so $|C|^2 = \frac{1}{2\pi\hbar}$. We may take $C = \frac{1}{\sqrt{2\pi\hbar}}$. This gives

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}. \quad (68)$$

Therefore,

$$\psi(x) = \int dp \langle x|p\rangle \langle p|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int dp e^{ipx/\hbar} \tilde{\psi}(p), \quad (69)$$

$$\tilde{\psi}(p) = \int dx \langle p|x\rangle \langle x|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int dx e^{-ipx/\hbar} \psi(x). \quad (70)$$

4.3 Translation Operator

4.3.1 Definition

$$T(a)|x\rangle = |x+a\rangle \quad (71)$$

To observe the effect of $T(a)$ on a wavefunction, consider

$$\begin{aligned}\langle x|T(a)|\psi\rangle &= \int dy \langle x|T(a)|y\rangle \langle y|\psi\rangle \\ &= \int dy \langle x|y+a\rangle \langle y|\psi\rangle \\ &= \int dy \delta(x-y-a) \psi(y) \\ &= \psi(x-a).\end{aligned}$$

4.3.2 Properties

- $T(0) = \mathbb{1}$;

- $T(a)T(b) = T(a+b) = T(b)T(a)$;
- $T(a)^{-1} = T(-a)$;
- $T(a)^\dagger = T(a)^{-1}$ (unitarity).

Proof. It is easy to see that

$$\langle y|T(a)^\dagger|x\rangle = \delta(y+a-x) = \langle y|T(-a)|x\rangle,$$

so $T(a)^\dagger = T(-a) = T(a)^{-1}$. □

4.4 Momentum as the Generator of Spatial Translation

$$T(a) = e^{-ia\hat{p}/\hbar} \equiv \sum_n \frac{(-ia\hat{p}/\hbar)^n}{n!} \quad (72)$$

Proof.

$$\begin{aligned} \langle x|e^{-ia\hat{p}/\hbar}|\psi\rangle &= \langle x|\sum_n \frac{(-ia\hat{p}/\hbar)^n}{n!}|\psi\rangle \\ &= \sum_n \frac{(-ia/\hbar)^n}{n!} \langle x|\hat{p}^n|\psi\rangle \\ &= \sum_n \frac{(-ia/\hbar)^n}{n!} \left(-i\hbar \frac{d}{dx}\right)^n \psi(x) \\ &= \sum_n \frac{1}{n!} \left(-a \frac{d}{dx}\right)^n \psi(x) \\ &= \psi(x-a) = \langle x|T(a)|\psi\rangle. \end{aligned}$$

□

4.5 Generalization to Three Spatial Dimensions

There is basically no new stuff in this section:

$$\mathbb{1} = \int d^3x |\mathbf{x}\rangle\langle\mathbf{x}|, \quad (73)$$

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}, \quad (74)$$

$$\langle \mathbf{x}|\hat{p}_i|\mathbf{y}\rangle = -i\hbar \frac{\partial}{\partial x_i} \delta^3(\mathbf{x}-\mathbf{y}), \quad \langle \mathbf{x}|\hat{\mathbf{p}}|\mathbf{y}\rangle = -i\hbar \nabla \delta^3(\mathbf{x}-\mathbf{y}), \quad (75)$$

$$\langle \mathbf{x}|\mathbf{y}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}, \quad (76)$$

$$T(\mathbf{a})|\mathbf{x}\rangle = |\mathbf{x} + \mathbf{a}\rangle, \quad (77)$$

$$T(\mathbf{a}) = e^{-i\mathbf{a}\cdot\mathbf{p}/\hbar}. \quad (78)$$

Shaojun Zan

5 Dynamics

5.1 The Schrödinger Equation & Stationary States

Assume that the Hamiltonian doesn't depend on time directly: $\frac{\partial}{\partial t} H = 0$.

We can then expand $|\psi, t\rangle$ using energy orthonormal basis, $H|n\rangle = E_n|n\rangle$:

$$|\psi, t\rangle = \sum_n |n\rangle \langle n|\psi, t\rangle \equiv \sum_n c_n(t) |n\rangle.$$

With the Schrödinger equation,

$$i\hbar \frac{\partial}{\partial t} |\psi, t\rangle = H |\psi, t\rangle, \quad (79)$$

$$i\hbar \frac{\partial}{\partial t} \left(\sum_n c_n(t) |n\rangle \right) = H \left(\sum_n c_n(t) |n\rangle \right) \Rightarrow \sum_n i\hbar \frac{dc_n}{dt} |n\rangle = \sum_n c_n(t) E_n |n\rangle.$$

Matching every term gives a differential equation for every n :

$$i\hbar \frac{dc_n}{dt} = E_n c_n \Rightarrow c_n(t) = c_n(t_0) e^{-iE_n(t-t_0)/\hbar}.$$

Thus,

$$|\psi, t\rangle = \sum_n c_n(t_0) e^{-iE_n(t-t_0)/\hbar} |n\rangle. \quad (80)$$

If a system starts with an energy eigenstate, then it stays in that eigenstate $\forall t \Rightarrow$ “**stationary state**”.

5.2 Time Evolution Operator

5.2.1 Definition

$$|\psi, t\rangle = U(t, t_0) |\psi, t_0\rangle \quad (81)$$

If $\frac{\partial}{\partial t} H = 0$, $U(t, t_0) = e^{-iH(t-t_0)/\hbar} \Rightarrow [H, U(t, t_0)] = 0$. It's easy to check why this is the case:

$$\frac{d}{dt} U(t, t_0) = -\frac{iH}{\hbar} U(t, t_0),$$

and rearrangement gives the Schrödinger equation for the operator U ,

$$i\hbar \frac{d}{dt} U(t, t_0) = H U(t, t_0).$$

This agrees with our previous Schrödinger equation by right multiplying a constant ket $|\psi, t_0\rangle$.

5.2.2 Properties

- $U(t_0, t_0) = 1$;
- $U(t, t_1)U(t_1, t_0) = U(t, t_0)$;
- $U(t, t_0)^\dagger = U(t, t_0)^{-1}$.

By this last property, we see that

$$\langle \psi, t | \psi, t \rangle = \langle U(t, t_0) | \psi, t_0 \rangle \langle U(t, t_0) | \psi, t_0 \rangle = \langle \psi, t_0 | U(t, t_0)^\dagger U(t, t_0) | \psi, t_0 \rangle = \langle \psi, t_0 | \psi, t_0 \rangle,$$

so a normalized state without measurements stays normalized.

5.3 Time Dependence of Expectation Values

$$\frac{d}{dt} \langle A, t \rangle = \frac{d}{dt} \langle \psi, t | A | \psi, t \rangle = \left(\frac{d}{dt} \langle \psi, t | \right) A | \psi, t \rangle + \langle \psi, t | A \left(\frac{d}{dt} | \psi, t \rangle \right)$$

Now $i\hbar \frac{d}{dt} | \psi, t \rangle = H | \psi, t \rangle$, and dual correspondence gives

$$-i\hbar \frac{d}{dt} \langle \psi, t | = \langle \psi, t | H.$$

Therefore,

$$\frac{d}{dt} \langle A, t \rangle = -\frac{1}{i\hbar} \langle \psi, t | H A | \psi, t \rangle + \frac{1}{i\hbar} \langle \psi, t | A H | \psi, t \rangle = \frac{1}{i\hbar} \langle \psi, t | [A, H] | \psi, t \rangle.$$

So,

$$\frac{d}{dt} \langle A, t \rangle = \frac{1}{i\hbar} \langle [A, H] \rangle. \quad (82)$$

Now

$$\langle A, t \rangle = \langle \psi, t | A | \psi, t \rangle = \langle \psi, t_0 | U(t, t_0)^\dagger A U(t, t_0) | \psi, t_0 \rangle,$$

when $[A, H] = 0$, $[A, U] = [A, e^{-iH\Delta t/\hbar}] = 0$,

$$\langle A, t \rangle = \langle \psi, t_0 | U(t, t_0)^\dagger U(t, t_0) A | \psi, t_0 \rangle = \langle A, t_0 \rangle.$$

5.4 Schrödinger, Heisenberg, and Interaction Pictures

	States	Operators
Schrödinger	time dependent	time independent
Heisenberg	time independent	time dependent

With fiducial time t_0 ,

$$|\psi\rangle_H = |\psi, t_0\rangle_S, \quad (83)$$

$$A_H = U(t, t_0)^\dagger A_S U(t, t_0). \quad (84)$$

5.4.1 Heisenberg Equation of Motion

We need $[H, U] = 0$ in order to reach the conclusion.

$$\begin{aligned} \frac{d}{dt} A_H &= \left(\frac{d}{dt} U^\dagger \right) A_S U + U^\dagger A_S \left(\frac{d}{dt} U \right) \\ &= \frac{i}{\hbar} (U^\dagger H A_S U - U^\dagger A_S H U) \\ &= \frac{1}{i\hbar} (U^\dagger A_S U U^\dagger H U - U^\dagger H U U^\dagger A_S U) \\ &= \frac{1}{i\hbar} [A_H, H] \end{aligned}$$

with $H_H = H_S \equiv H$, $\partial_t H = 0 \Rightarrow [H, U] = 0$. Thus,

$$\frac{d}{dt} A_H = \frac{1}{i\hbar} [A_H, H]. \quad (85)$$

The solution to this equation is

$$A_H = e^{iHt/\hbar} A_S e^{-iHt/\hbar}. \quad (86)$$

5.4.2 The Interaction Picture

Let $H = H^0 + H^1$ with solvable H^0 and perturbation H^1 . Beginning with the Schrödinger picture, the interaction picture defines

$$|\psi, t\rangle_I = e^{iH_S^0 t/\hbar} |\psi, t\rangle_S, \quad (87)$$

$$A_I = e^{iH_S^0 t/\hbar} A_S e^{-iH_S^0 t/\hbar}. \quad (88)$$

So, the interaction picture coincides with the Heisenberg picture when the perturbation $H^1 = 0$.

To derive the equation of motion in the interaction picture,

$$\begin{aligned} i\hbar \frac{d}{dt} |\psi, t\rangle_I &= i\hbar \frac{d}{dt} \left(e^{iH_S^0 t/\hbar} |\psi, t\rangle_S \right) \\ &= i\hbar \frac{i}{\hbar} H_S^0 e^{iH_S^0 t/\hbar} |\psi, t\rangle_S + e^{iH_S^0 t/\hbar} H_S |\psi, t\rangle_S \\ &= e^{iH_S^0 t/\hbar} H_S^1 |\psi, t\rangle_S \\ &= e^{iH_S^0 t/\hbar} H_S^1 e^{-iH_S^0 t/\hbar} |\psi, t\rangle_I \\ &\equiv H_I^1 |\psi, t\rangle_I. \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{d}{dt} A_I &= \frac{d}{dt} \left(e^{iH_S^0 t/\hbar} A_S e^{-iH_S^0 t/\hbar} \right) \\ &= \frac{i}{\hbar} H_S^0 e^{iH_S^0 t/\hbar} A_S e^{-iH_S^0 t/\hbar} - \frac{i}{\hbar} e^{iH_S^0 t/\hbar} A_S H_S^0 e^{-iH_S^0 t/\hbar} \\ &= \frac{1}{i\hbar} [A_I, H_S^0]. \end{aligned}$$

Now $H_I^0 = e^{iH_S^0 t/\hbar} H_S^0 e^{-iH_S^0 t/\hbar} = H_S^0$ (not necessarily for H_I^1),

$$\frac{d}{dt} A_I = \frac{1}{i\hbar} [A_I, H_I^0]. \quad (89)$$

5.5 Ehrenfest Relations

With $H = \frac{\mathbf{p}^2}{2m} + V(\mathbf{x})$,

$$\frac{d}{dt} \mathbf{p} = \frac{1}{i\hbar} [\mathbf{p}, H] = -\nabla V, \quad (90)$$

$$\frac{d}{dt} \mathbf{x} = \frac{1}{i\hbar} [\mathbf{x}, H] = \frac{1}{m} \mathbf{p}. \quad (91)$$

6 The Simple Harmonic Oscillator & Coherent States

6.1 SHO Spectrum

With $m = \omega = \hbar = 1$,

$$H = \frac{1}{2} (\hat{p}^2 + \hat{x}^2), \quad (92)$$

$$[\hat{x}, \hat{p}] = i. \quad (93)$$

6.1.1 Ladder Operators

$$a = \frac{1}{\sqrt{2}} (\hat{x} + i\hat{p}), \quad a^\dagger = \frac{1}{\sqrt{2}} (\hat{x} - i\hat{p}), \quad (94)$$

$$\hat{x} = \frac{1}{\sqrt{2}} (a + a^\dagger), \quad \hat{p} = \frac{-i}{\sqrt{2}} (a - a^\dagger). \quad (95)$$

6.1.2 Corollary

- $[a, a^\dagger] = 1$;

We may see easily that

$$[a, a^\dagger] = \frac{1}{2} [\hat{x} + i\hat{p}, \hat{x} - i\hat{p}] = \frac{1}{2} (-i \cdot i + i \cdot -i) = 1.$$

- $H = a^\dagger a + \frac{1}{2} \equiv N + \frac{1}{2}$.

$$\begin{aligned} a^\dagger a &= \frac{1}{2} (\hat{x} - i\hat{p}) (\hat{x} + i\hat{p}) \\ &= \frac{1}{2} (\hat{x}^2 - i\hat{p}\hat{x} + i\hat{x}\hat{p} + \hat{p}^2) \\ &= \frac{1}{2} (\hat{p}^2 + \hat{x}^2 - 1). \end{aligned}$$

6.1.3 Raising and Lowering Operators

It is easy to see

$$[N, a] = [a^\dagger a, a] = [a^\dagger, a] a = -a, \quad (96)$$

$$[N, a^\dagger] = [a^\dagger a, a^\dagger] = a^\dagger [a, a^\dagger] = a^\dagger. \quad (97)$$

Now $\langle \phi | N | \phi \rangle = \langle \phi | a^\dagger a | \phi \rangle = \|a | \phi \rangle\|^2 \geq 0$,

$$N | \nu \rangle = \nu | \nu \rangle \quad \Rightarrow \quad \nu \geq 0.$$

$$\begin{cases} Na|\nu\rangle = (-a + aN)|\nu\rangle = (\nu - 1)a|\nu\rangle & \text{lowering operator} \\ Na^\dagger|\nu\rangle = (a^\dagger + a^\dagger N)|\nu\rangle = (\nu + 1)a^\dagger|\nu\rangle & \text{raising operator} \end{cases} \quad (98)$$

6.1.4 Nonnegative Integer Eigenvalues

Positivity requires $\|a|\nu\rangle\|^2 = \nu \geq 0$ and $\|a^\dagger|\nu\rangle\|^2 = \nu + 1 \geq 0$, which then give $\nu \geq 0$. Hence, $a|\nu\rangle = 0 \Leftrightarrow \nu = 0$.

Now consider non-integer $\nu = 1/2$. $a|1/2\rangle \neq 0$ and leads to an eigenvalue of $-1/2$ for the lowered state, contradicting $\nu \geq 0$.

6.1.5 Corollary

Substituting ν by n , $\|a|n\rangle\|^2 = n$ and $\|a^\dagger|n\rangle\|^2 = n + 1$ yield

$$a|n\rangle = \sqrt{n}|n-1\rangle, \quad a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle. \quad (99)$$

Also, $a|0\rangle = 0$, and

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle. \quad (100)$$

6.2 SHO in Heisenberg Picture

By $H = \frac{1}{2}(\hat{p}^2 + \hat{x}^2)$,

$$\begin{aligned} \frac{d}{dt}\hat{p} &= -i[\hat{p}, H] = -\frac{i}{2}[\hat{p}, \hat{x}^2] = -\hat{x}, \\ \frac{d}{dt}\hat{x} &= -i[\hat{x}, H] = -\frac{i}{2}[\hat{x}, \hat{p}^2] = \hat{p}. \end{aligned}$$

Written in the matrix form,

$$\frac{d}{dt} \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix} = i\sigma_2 \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix} \Rightarrow \begin{pmatrix} \hat{x} \\ \hat{p} \end{pmatrix} = \begin{pmatrix} \hat{x}_0 \\ \hat{p}_0 \end{pmatrix} e^{i\sigma_2 t},$$

where σ_2 is the Pauli matrix:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (101)$$

Defining the matrix exponential is achieved via the Taylor expansion:

$$e^A \equiv \sum_{n=0}^{\infty} \frac{1}{n!} A^n. \quad (102)$$

So,

$$\begin{aligned}
e^{i\sigma_2 t} &= \sum_{n=0}^{\infty} \frac{(i\sigma_2 t)^n}{n!} \\
&= \sum_{n=0}^{\infty} \left[\frac{(i\sigma_2 t)^{2n}}{(2n)!} + \frac{(i\sigma_2 t)^{2n+1}}{(2n+1)!} \right] \\
&= \sum_{n=0}^{\infty} \left[\mathbb{1}(-1)^n \frac{t^{2n}}{(2n)!} + i\sigma_2(-1)^n \frac{t^{2n+1}}{(2n+1)!} \right] \\
&= \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} \\
&= R(-t),
\end{aligned}$$

which is actually clockwise rotation (CW):

$$e^{i\sigma_2 t} = \cos(\sigma_2 t) + i \sin(\sigma_2 t) = \mathbb{1} \cos t + i\sigma_2 \sin t. \quad (103)$$

6.3 Coherent / Glauber States

6.3.1 Definition

- $a|\alpha\rangle = \alpha|\alpha\rangle$;

We see very clearly that $\alpha \in \mathbb{C}$, as $a \neq a^\dagger$.

- $\langle\alpha|\alpha\rangle = 1$.

$$\langle E \rangle = \langle\alpha|H|\alpha\rangle = \langle\alpha|\left(a^\dagger a + \frac{1}{2}\right)|\alpha\rangle = |\alpha|^2 + \frac{1}{2}$$

6.3.2 Displacement Operator

$$D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a) \quad (104)$$

6.3.3 Hadamard's Lemma

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2!}[A, [A, B]] + \dots \quad (105)$$

Proof. Let $F(s) = e^{sA} B e^{-sA}$. Then,

$$F'(s) = A e^{sA} B e^{-sA} - e^{sA} B A e^{-sA}.$$

Now, as $[A, e^{sA}] = 0$,

$$F'(s) = e^{sA} (AB - BA) e^{-sA} = e^{sA} [A, B] e^{-sA}.$$

Similarly,

$$F''(s) = e^{sA} A (AB - BA) e^{-sA} - e^{sA} (AB - BA) A e^{-sA} = e^{sA} [A, [A, B]] e^{-sA}.$$

Hence, by Taylor expansion,

$$F(s) = F(0) + F'(0)s + \frac{1}{2!} F''(0)s^2 + \dots, \quad (106)$$

at $s = 1$,

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2!} [A, [A, B]] + \dots \quad (107)$$

□

Another piece of corollary related to the commutators is

$$[A, [A, B]] = [B, [A, B]] = 0 \quad \Rightarrow \quad e^A e^B = e^{A+B+[A,B]/2} = e^{A+B} e^{[A,B]/2}. \quad (108)$$

Proof. Let $F(s) = e^{sA} e^{sB}$. Thus,

$$F'(s) = A e^{sA} e^{sB} + e^{sA} B e^{sB}.$$

Now $[A, [A, B]] = 0$, we know from Hadamard's lemma that

$$e^{sA} B e^{-sA} = B + s[A, B].$$

$$\begin{aligned} F'(s) &= A e^{sA} e^{sB} + e^{sA} B e^{-sA} e^{sA} e^{sB} \\ &= (A + B + s[A, B]) e^{sA} e^{sB} \\ &\equiv X(s) F(s) \end{aligned}$$

Also, as $[B, [A, B]] = 0$, $e^{-sB} A e^{sB} = A + e[-B, A] = A + s[A, B]$, and

$$\begin{aligned} F'(s) &= e^{sA} e^{sB} e^{-sB} A e^{sB} + e^{sA} e^{sB} B \\ &= e^{sA} e^{sB} (A + B + s[A, B]) \\ &\equiv F(s) X(s). \end{aligned}$$

We've thus arrived at a differential equation: $dF/F = X ds$, and therefore,

$$F(s) = N \exp \left(As + Bs + \frac{1}{2} [A, B] s^2 \right).$$

For $s = 1$, $N = 1$,

$$e^A e^B = e^{A+B+[A,B]/2} = e^{A+B} e^{[A,B]/2} \quad (109)$$

as $[A, [A, B]] = [B, [A, B]] = 0$. Accordingly,

$$e^B e^A = e^{A+B+[B,A]/2} = e^{A+B-[A,B]/2}, \quad (110)$$

$$e^A e^B = e^B e^A e^{[A,B]}. \quad (111)$$

□

6.3.4 Properties

- $D(\alpha)^\dagger = D(\alpha)^{-1} = D(-\alpha)$;

Proof. As we've known,

$$D(\alpha) = e^{\alpha a^\dagger - \alpha^* a} = e^{-|\alpha|^2/2} e^{\alpha a^\dagger} e^{-\alpha^* a} = e^{|\alpha|^2/2} e^{-\alpha^* a} e^{\alpha a^\dagger}, \quad (112)$$

where the new expressions come from the fact that a and $a^\dagger$ both commute with $[a, a^\dagger]$. So,

$$D(\alpha) D(-\alpha) = e^{|\alpha|^2/2} e^{-\alpha^* a} e^{\alpha a^\dagger} e^{-|\alpha|^2/2} e^{-\alpha a^\dagger} e^{\alpha^* a} = \mathbb{1}$$

trivially. Thus, $D(-\alpha) = D(\alpha)^{-1}$. Also,

$$\begin{aligned} D(\alpha)^\dagger D(\alpha) &= e^{|\alpha|^2/2} \left(e^{-\alpha^* a} e^{\alpha a^\dagger} \right)^\dagger e^{-|\alpha|^2/2} e^{\alpha a^\dagger} e^{-\alpha^* a} \\ &= \left(e^{\alpha a^\dagger} \right)^\dagger \left(e^{-\alpha^* a} \right)^\dagger e^{\alpha a^\dagger} e^{-\alpha^* a} \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left[(\alpha a^\dagger)^n \right]^\dagger \sum_{m=0}^{\infty} \frac{1}{m!} [(-\alpha^* a)^m]^\dagger e^{\alpha a^\dagger} e^{-\alpha^* a} \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (\alpha^* a)^n \sum_{m=0}^{\infty} \frac{1}{m!} (-\alpha a^\dagger)^m e^{\alpha a^\dagger} e^{-\alpha^* a} \\ &= e^{\alpha^* a} e^{-\alpha a^\dagger} e^{\alpha a^\dagger} e^{-\alpha^* a} \\ &= \mathbb{1}, \end{aligned}$$

so $D(\alpha)^\dagger = D(\alpha)^{-1}$. □

- $D(\alpha)^\dagger a D(\alpha) = a + \alpha$;

Proof. From Hadamard's lemma,

$$\begin{aligned} D(\alpha)^\dagger a D(\alpha) &= e^A a e^{-A} \quad \text{with} \quad A = \alpha^* a - \alpha a^\dagger \\ &= a + [A, a] + \frac{1}{2!}[A, [A, a]] + \dots \end{aligned}$$

Now $[A, a] = -\alpha[a^\dagger, a] = \alpha$, $[A, [A, a]] = 0$, so $D(\alpha)^\dagger a D(\alpha) = a + \alpha$.

□

- $D(\alpha + \beta) = D(\alpha)D(\beta)e^{-\text{Im}(\alpha\beta^*)}$.

Proof. Firstly,

$$D(\alpha + \beta) = e^{|\alpha + \beta|^2/2} e^{-(\alpha^* + \beta^*)a} e^{(\alpha + \beta)a^\dagger},$$

where

$$\begin{aligned} |\alpha + \beta|^2 &= (\alpha + \beta)(\alpha + \beta)^* \\ &= |\alpha|^2 + |\beta|^2 + \alpha^* \beta + \alpha \beta^* \\ &= |\alpha|^2 + |\beta|^2 + [\alpha \beta^* + (\alpha \beta^*)^*] \\ &= |\alpha|^2 + |\beta|^2 + 2\text{Re}(\alpha \beta^*). \end{aligned}$$

As for the rest, $e^{-(\alpha^* + \beta^*)a} e^{(\alpha + \beta)a^\dagger} = e^{-\alpha^* a} e^{-\beta^* a} e^{\alpha a^\dagger} e^{\beta a^\dagger}$, by $e^A e^B = e^B e^A e^{[A, B]}$,

$$e^{-\alpha^* a} \left(e^{-\beta^* a} e^{\alpha a^\dagger} \right) e^{\beta a^\dagger} = e^{-\alpha^* a} \left(e^{\alpha a^\dagger} e^{-\beta^* a} e^{[-\beta^* a, \alpha a^\dagger]} \right) e^{\beta a^\dagger}.$$

As $[-\beta^* a, \alpha a^\dagger] = -\alpha \beta^* = -\text{Re}(\alpha \beta^*) - i\text{Im}(\alpha \beta^*)$,

$$\begin{aligned} D(\alpha + \beta) &= e^{|\alpha + \beta|^2/2} e^{-(\alpha^* + \beta^*)a} e^{(\alpha + \beta)a^\dagger} \\ &= e^{|\alpha|^2/2} e^{|\beta|^2/2} e^{-\alpha^* a} e^{\alpha a^\dagger} e^{-\beta^* a} e^{\beta a^\dagger} e^{-i\text{Im}(\alpha \beta^*)} \\ &= D(\alpha)D(\beta)e^{-i\text{Im}(\alpha \beta^*)}. \end{aligned}$$

□

6.3.5 Theorem of Displacement Operator

$$|\alpha\rangle = D(\alpha) |0\rangle \quad (113)$$

Proof. We want to prove that $a|\alpha\rangle = \alpha|\alpha\rangle$:

$$\begin{aligned} aD(\alpha) |0\rangle &= D(\alpha) [D(\alpha)^\dagger a D(\alpha)] |0\rangle \\ &= D(\alpha) (a + \alpha) |0\rangle \\ &= \alpha D(\alpha) |0\rangle. \end{aligned}$$

It is also necessary to prove uniqueness, by which we mean that $|0\rangle$ is unique with $a|0\rangle = 0$. So,

$$aD(-\alpha) |\alpha\rangle = D(-\alpha) D(-\alpha)^\dagger a D(-\alpha) |\alpha\rangle = D(-\alpha) (a - \alpha) |\alpha\rangle = 0.$$

□

With this theorem, we know that

$$\begin{aligned} |\alpha\rangle &= D(\alpha) |0\rangle \\ &= e^{-|\alpha|^2/2} e^{\alpha a^\dagger} e^{-\alpha^* a} |0\rangle \\ &= e^{-|\alpha|^2/2} e^{\alpha a^\dagger} |0\rangle \\ &= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{1}{n!} (\alpha a^\dagger)^n |0\rangle \\ &= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \left[\frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle \right], \\ |\alpha\rangle &= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \end{aligned} \quad (114)$$

The probability of measuring the n -th state is:

$$P_n = |\langle n|\alpha\rangle|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}, \quad (115)$$

which turns out to conform with the Poisson distribution.

6.3.6 Inner Product

$$\begin{aligned}
\langle \beta | \alpha \rangle &= \langle 0 | D(\beta)^\dagger D(\alpha) | 0 \rangle \\
&= \langle 0 | D(-\beta) D(\alpha) | 0 \rangle \\
&= \langle 0 | D(\alpha - \beta) | 0 \rangle e^{-i\text{Im}(\alpha^* \beta)} \\
&= e^{-|\alpha - \beta|^2/2} \langle 0 | e^{(\alpha - \beta)a^\dagger} e^{-(\alpha^* - \beta^*)a} | 0 \rangle e^{i\text{Im}(\alpha \beta^*)} \\
&= e^{-|\alpha - \beta|^2/2} e^{i\text{Im}(\alpha \beta^*)} \\
&= e^{-|\alpha|^2/2} e^{-|\beta|^2/2} e^{\text{Re}(\alpha \beta^*)} e^{i\text{Im}(\alpha \beta^*)} \\
&= e^{-|\alpha|^2/2} e^{-|\beta|^2/2} e^{\alpha \beta^*}.
\end{aligned}$$

With this in mind, we may calculate the modulus squared:

$$\begin{aligned}
|\langle \beta | \alpha \rangle|^2 &= \left(e^{-|\alpha - \beta|^2/2} e^{i\text{Im}(\alpha \beta^*)} \right) \left(e^{-|\alpha - \beta|^2/2} e^{i\text{Im}(\alpha \beta^*)} \right)^* \\
&= e^{-|\alpha - \beta|^2} e^{i\text{Im}(\alpha \beta^*)} e^{-i\text{Im}(\alpha \beta^*)} \\
&= e^{-|\alpha - \beta|^2} \neq 0 \neq 1,
\end{aligned}$$

which means that the states are not orthogonal.

6.3.7 Over-Completeness

$$\mathbb{1} = \frac{1}{\pi} \int d^2\alpha |\alpha\rangle\langle\alpha| \tag{116}$$

Proof. In order to see this, let us remember that

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle,$$

$$\langle\alpha| = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^{*n}}{\sqrt{n!}} \langle n|,$$

and thus

$$|\alpha\rangle\langle\alpha| = e^{-|\alpha|^2} \sum_{nm} \frac{\alpha^{*m} \alpha^n}{\sqrt{n!m!}} |n\rangle\langle m|.$$

Now, as $\int_0^\infty dt t^n e^{-t} = n!$, and in polar coordinates, $\alpha = r e^{i\theta}$, $\alpha^* = r e^{-i\theta}$,

$$\begin{aligned} \int d^2\alpha e^{-|\alpha|^2} \alpha^{*m} \alpha^n &= \int dr d\theta r e^{-r^2} r^m e^{-im\theta} r^n e^{in\theta} \\ &= \frac{1}{2} \int dr^2 e^{-r^2} r^{m+n} \int d\theta e^{i(n-m)\theta} \\ &= \pi \delta_{nm} \left(\frac{m+n}{2} \right)!. \end{aligned}$$

Therefore,

$$\begin{aligned} \int d^2\alpha |\alpha\rangle\langle\alpha| &= \sum_{nm} \int d^2\alpha e^{-|\alpha|^2} \alpha^{*m} \alpha^n \frac{1}{\sqrt{n!m!}} |n\rangle\langle m| \\ &= \sum_{nm} \pi \delta_{nm} \left(\frac{m+n}{2} \right)! \frac{1}{\sqrt{n!m!}} |n\rangle\langle m| \\ &= \pi \sum_n |n\rangle\langle n| = \pi \mathbb{1}. \end{aligned}$$

Thus,

$$\mathbb{1} = \frac{1}{\pi} \int d^2\alpha |\alpha\rangle\langle\alpha|.$$

□

With this resolution of identity, we can deconstruct a coherent state:

$$|\beta\rangle = \frac{1}{\pi} \int d^2\alpha |\alpha\rangle\langle\alpha|\beta\rangle = \frac{1}{\pi} \int d^2\alpha |\alpha\rangle e^{-|\alpha|^2/2} e^{-|\beta|^2/2} e^{\beta\alpha^*}, \quad (117)$$

where we can see again that the states are not linearly independent.

6.3.8 Balanced Minimal Uncertainty States

As we know from coherent states,

$$\begin{aligned} \langle x \rangle &= \frac{1}{\sqrt{2}} \langle \alpha | (a + a^\dagger) | \alpha \rangle = \frac{1}{\sqrt{2}} (\alpha + \alpha^*) = \sqrt{2} \operatorname{Re}(\alpha), \\ \langle p \rangle &= \frac{-i}{\sqrt{2}} \langle \alpha | (a - a^\dagger) | \alpha \rangle = \frac{-i}{\sqrt{2}} (\alpha - \alpha^*) = \sqrt{2} \operatorname{Im}(\alpha). \end{aligned}$$

Further calculation easily shows that

$$\Delta x = \Delta p = \frac{1}{\sqrt{2}},$$

which explains the balance, and

$$\Delta x \Delta p = \frac{1}{2},$$

which explains the minimum.

6.3.9 Time Evolution

Let $|\psi, t = 0\rangle = |\alpha_0\rangle$. Then,

$$\begin{aligned}
 |\psi, t\rangle &= e^{-iHt} e^{-|\alpha_0|^2/2} \sum_n \frac{\alpha_0^n}{\sqrt{n!}} |n\rangle \\
 &= e^{-|\alpha_0|^2/2} \sum_n \frac{\alpha_0^n}{\sqrt{n!}} e^{-iHt} |n\rangle \\
 &= e^{-|\alpha_0|^2/2} \sum_n \frac{\alpha_0^n}{\sqrt{n!}} e^{-i(n+1/2)t} |n\rangle \\
 &= e^{-it/2} e^{-|\alpha_0|^2/2} \sum_n \frac{\alpha^n(t)}{\sqrt{n!}} |n\rangle \\
 &= e^{-it/2} |\alpha(t)\rangle,
 \end{aligned}$$

with $\alpha(t) = \alpha_0 e^{-it}$.

Shaojun Zan

7 Symmetry and Angular Momentum

7.1 Symmetry in Quantum Mechanics

Consider unitary transformation $B|\psi\rangle = |\psi'\rangle$. This implies us an operator O in new basis is given by:

$$\begin{aligned} O' &= B O B^\dagger, \\ \text{new} \leftarrow \text{new} &= \text{new} \leftarrow \text{old} \leftarrow \text{old} \leftarrow \text{new}. \end{aligned} \quad (118)$$

This directly gives that

$$\langle\psi'|O'|\psi'\rangle = \langle\psi|B^\dagger B O B^\dagger B|\psi\rangle = \langle\psi|O|\psi\rangle, \quad (119)$$

so a change of basis cannot influence the expectation values.

We may also add time evolution: $|\psi', t\rangle = U(t, t_0)B|\psi, t_0\rangle$, where $[B, U] = 0$. Thus,

$$\begin{aligned} \langle\psi', t|O'|\psi', t\rangle &= \langle\psi, t_0|B^\dagger U^\dagger B O B^\dagger U B|\psi, t_0\rangle \\ &= \langle\psi, t_0|U^\dagger O U|\psi, t_0\rangle \\ &= \langle\psi, t|O|\psi, t\rangle. \end{aligned}$$

If observable B satisfies $[B, U] = 0 \ \forall t$, then B corresponds to a symmetry of the quantum system. This is clearly a particular example of Noether's theorem. Additionally,

$$[B, U] = 0 \quad \Leftrightarrow \quad [B, H] = 0. \quad (120)$$

This is easily proved when $\partial_t H = 0 \Rightarrow U = e^{-iHt/\hbar}$:

$$\frac{d}{dt} [B, e^{-iHt/\hbar}] = \left[B, -\frac{iH}{\hbar} e^{-iHt/\hbar} \right] = 0$$

and let $t = 0$, so $[B, H] = 0$.

7.2 Translational Symmetry

We know that translation is generated via the operator $T(\mathbf{a}) = e^{-i\mathbf{a}\cdot\mathbf{p}/\hbar}$. Therefore, translational symmetry gives

$$[T(\mathbf{a}), H] = [e^{-i\mathbf{a}\cdot\mathbf{p}/\hbar}, H] = 0.$$

Differentiate by $\mathbf{a}$ and take $\mathbf{a} = 0$, we see that $[\mathbf{p}, H] = 0$, i.e. momentum is conserved.

7.3 Rotational Symmetry

As $T(\mathbf{a})|\mathbf{x}\rangle = |\mathbf{x} + \mathbf{a}\rangle$, consider

$$|\mathbf{x}'\rangle = \mathcal{R}(\mathbf{n}, \omega)|\mathbf{x}\rangle = |R(\mathbf{n}, \omega)\mathbf{x}\rangle \quad (121)$$

with unit vector $\mathbf{n}$ and angle ω .

Now the unit vector $\mathbf{n} = \mathbf{n} = 1$ may be specified by 2 angles (azimuth and zenith), so **a general rotation can be specified by 3 angles in total.**

We may write

$$R(\mathbf{n}, \omega) \equiv R(n_1 \hat{\mathbf{e}}_1, \omega_1) R(n_2 \hat{\mathbf{e}}_2, \omega_2) R(n_3 \hat{\mathbf{e}}_3, \omega_3), \quad (122)$$

where

$$R_1(\omega) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \omega & -\sin \omega \\ 0 & \sin \omega & \cos \omega \end{pmatrix} \Rightarrow N_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad (123)$$

$$R_2(\omega) = \begin{pmatrix} \cos \omega & 0 & -\sin \omega \\ 0 & 1 & 0 \\ \sin \omega & 0 & \cos \omega \end{pmatrix} \Rightarrow N_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad (124)$$

$$R_3(\omega) = \begin{pmatrix} \cos \omega & -\sin \omega & 0 \\ \sin \omega & \cos \omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow N_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (125)$$

$$(N_i)_{jk} = -\varepsilon_{ijk}. \quad (126)$$

The matrices N_i are called the **infinitesimal generators** of these rotations, derived by taking the derivative with respect to the angle ω and set $\omega = 0$ afterwards. Therefore, the action of the general rotation is given by

$$R(\mathbf{n}, \omega) = \prod_{i=1}^3 [1 + \omega n_i N_i + \mathcal{O}(\omega^2)] = 1 + \omega \mathbf{n} \cdot \mathbf{N} + \mathcal{O}(\omega^2). \quad (127)$$

Now we may consider the action of a rotation on the position vector $\mathbf{x}$:

$$\begin{aligned} [R(\mathbf{n}, \delta\omega) \mathbf{x}]_i &= [R(\mathbf{n}, \delta\omega)]_{ij} x_j \\ &\approx (1 + \delta\omega \mathbf{n} \cdot \mathbf{N})_{ij} x_j \\ &= [\delta_{ij} + \delta\omega n_k (N_k)_{ij}] x_j \\ &= x_i - \delta\omega \varepsilon_{kij} n_k x_j \\ &= x_i + \delta\omega (\mathbf{n} \times \mathbf{x})_i. \end{aligned}$$

Thus,

$$R(\mathbf{n}, \delta\omega) \mathbf{x} = \mathbf{x} + \delta\omega \mathbf{n} \times \mathbf{x} + \mathcal{O}(\delta\omega^2). \quad (128)$$

7.4 Rotations in Quantum Mechanics

As a reminder of our notations,

$$|\mathbf{x}'\rangle = \mathcal{R}(\mathbf{n}, \omega) |\mathbf{x}\rangle = |R(\mathbf{n}, \omega) \mathbf{x}\rangle.$$

7.4.1 Properties

- $R(\mathbf{n}, \omega = 0) = \mathbb{1}$;
- $R(\mathbf{n}, \omega)^{-1} = R(\mathbf{n}, \omega)^\dagger = R(\mathbf{n}, -\omega)$.

7.4.2 Rotated State

Let $|\psi'\rangle \equiv \mathcal{R}(\mathbf{n}, \omega) |\psi\rangle$. Then,

$$\psi'(x') = \langle \mathbf{x}' | \psi' \rangle = \langle \mathbf{x} | \mathcal{R}(\mathbf{n}, \omega)^\dagger \mathcal{R}(\mathbf{n}, \omega) | \psi \rangle = \langle \mathbf{x} | \psi \rangle = \psi(x),$$

$$\psi'(x) = \langle \mathbf{x} | \psi' \rangle = \langle \mathbf{x} | \mathcal{R}(\mathbf{n}, \omega) | \psi \rangle = \langle R(\mathbf{n}, -\omega) \mathbf{x} | \psi \rangle = \psi(R(\mathbf{n}, -\omega) \mathbf{x}).$$

From our conclusion in the last subsection, for infinitesimal $\delta\omega$,

$$\psi'(\mathbf{x}) = \psi(R(\mathbf{n}, -\delta\omega) \mathbf{x}) = \psi(\mathbf{x} - \delta\omega \mathbf{n} \times \mathbf{x}).$$

7.4.3 Angular Momentum as the Generator of Rotations

We define

$$R(\mathbf{n}, \omega) = e^{-i\omega \mathbf{n} \cdot \mathbf{L} / \hbar}, \quad (129)$$

where $\mathbf{L} = \mathbf{x} \times \mathbf{p}$. The operator $\mathbf{L}$ is clearly Hermitian, as

$$\mathbf{L} = L_i \hat{\mathbf{e}}_i = \varepsilon_{ijk} x_j p_k \hat{\mathbf{e}}_i \Rightarrow \mathbf{L}^\dagger = \mathbf{L}.$$

The commutation relations are given by

$$[L_i, L_j] = i\hbar \varepsilon_{ijk} L_k. \quad (130)$$

We may prove this definition of rotation operator:

$$\begin{aligned}
\langle \mathbf{x} | e^{-i\delta\omega \mathbf{n} \cdot \mathbf{L} / \hbar} | \psi \rangle &\approx \langle \mathbf{x} | \left(\mathbb{1} - \frac{i\delta\omega}{\hbar} \mathbf{n} \cdot \mathbf{L} \right) | \psi \rangle \\
&= \langle \mathbf{x} | \left[\mathbb{1} - \frac{i\delta\omega}{\hbar} \mathbf{n} \cdot (\mathbf{x} \times \mathbf{p}) \right] | \psi \rangle \\
&= \langle \mathbf{x} | \psi \rangle - \frac{i\delta\omega}{\hbar} \langle \mathbf{x} | \varepsilon_{ijk} n_i x_j p_k | \psi \rangle \\
&= \langle \mathbf{x} | \psi \rangle - \frac{i\delta\omega}{\hbar} \varepsilon_{ijk} n_i x_j \cdot \left(-i\hbar \frac{\partial}{\partial x_k} \langle \mathbf{x} | \psi \rangle \right) \\
&= \langle \mathbf{x} | \psi \rangle - \delta\omega \varepsilon_{ijk} n_i x_j \frac{\partial}{\partial x_k} \langle \mathbf{x} | \psi \rangle \\
&= \langle \mathbf{x} | \psi \rangle - \delta\omega (\mathbf{n} \times \mathbf{x}) \cdot \nabla \langle \mathbf{x} | \psi \rangle \\
&= \psi(\mathbf{x} - \delta\omega \mathbf{n} \times \mathbf{x}) \\
&= \langle \mathbf{x} | R(\mathbf{n}, \delta\omega) | \psi \rangle,
\end{aligned}$$

so it is true that

$$R(\mathbf{n}, \omega) = e^{-i\omega \mathbf{n} \cdot \mathbf{L} / \hbar}.$$

7.4.4 Rotational Symmetry

Rotational symmetry means $[\mathcal{R}(\mathbf{n}, \omega), H] = [e^{-i\omega \mathbf{n} \cdot \mathbf{L} / \hbar}, H] = 0$. Similarly, differentiate by ω and take $\omega = 0$, we get

$$[\mathbf{n} \cdot \mathbf{L}, H] = 0 \quad \Rightarrow \quad [\mathbf{L}, H] = 0. \quad (131)$$

7.5 Abstract Angular Momentum

Now we derive some properties of the angular momentum purely algebraically, via the commutator and the abstract notation J :

$$[J_i, J_j] = i\hbar \varepsilon_{ijk} J_k. \quad (132)$$

7.5.1 Simultaneous Eigenstates

The Casimir operator $J^2 = J_i J_i$ satisfies $[J^2, J_i] = 0$:

$$\begin{aligned}
[J^2, J_i] &= [J_k J_k, J_i] \\
&= J_k [J_k, J_i] + [J_k, J_i] J_k \\
&= i\hbar \varepsilon_{kil} J_k J_l + i\hbar \varepsilon_{kil} J_l J_k \\
&= i\hbar (\varepsilon_{kil} + \varepsilon_{lik}) J_k J_l = 0.
\end{aligned}$$

Therefore, it is possible to construct simultaneous and orthogonal eigenstates $\langle \lambda m | \lambda' m' \rangle = \delta_{\lambda\lambda'} \delta_{mm'}$, where

$$J^2 |\lambda m\rangle = \lambda \hbar^2 |\lambda m\rangle, \quad J_3 |\lambda m\rangle = m \hbar |\lambda m\rangle.$$

A trivial but subtle point is that we must have $\lambda > 0$, as $J^2 = J_i J_i = (J_i)^\dagger J_i$.

7.5.2 Raising and Lowering Operators

We define

$$J_\pm = J_1 \pm iJ_2. \quad (133)$$

- $J_\pm^\dagger = J_\mp$;

Proof. $(J_\pm)^\dagger = (J_1 \pm iJ_2)^\dagger = J_1 \mp iJ_2 = J_\mp.$ □

- $[J_3, J_\pm] = \pm \hbar J_\pm$;

Proof. By the commutation relation,

$$\begin{aligned} [J_3, J_\pm] &= [J_3, J_1] \pm i[J_3, J_2] \\ &= i\hbar J_2 \pm i(-i\hbar J_1) \\ &= \pm \hbar J_\pm. \end{aligned}$$

□

- $J_\mp J_\pm = J^2 - J_3 (J_3 \pm \hbar).$

Proof. Expanding the operators,

$$\begin{aligned} J_\mp J_\pm &= (J_1 \mp iJ_2)(J_1 \pm iJ_2) \\ &= J_1^2 \pm i[J_1, J_2] + J_2^2 \\ &= J_1^2 + J_2^2 \mp \hbar J_3 \\ &= J^2 - J_3^2 \mp \hbar J_3 \\ &= J^2 - J_3 (J_3 \pm \hbar). \end{aligned}$$

□

With all these properties, we see that

$$J_3 J_{\pm} |\lambda m\rangle = (\pm \hbar J_{\pm} + J_{\pm} J_3) |\lambda m\rangle = (m \pm 1) \hbar J_{\pm} |\lambda m\rangle,$$

which means that $J_{\pm} |\lambda m\rangle$ are eigenstates of J_3 with eigenvalues $m \pm 1$:

$$J_{\pm} |\lambda m\rangle = N_{\pm} |\lambda (m \pm 1)\rangle. \quad (134)$$

Our properties before help us find the constants $N_{\pm}$:

$$\begin{aligned} |N_{\pm}|^2 &= \langle \lambda m | J_{\mp} J_{\pm} | \lambda m \rangle \\ &= \langle \lambda m | [J^2 - J_3 (J_3 \pm \hbar)] | \lambda m \rangle \\ &= [\lambda - m(m \pm 1)] \hbar^2 \geq 0. \end{aligned}$$

Therefore, this tells us that $m(m \pm 1) \leq \lambda$, so m is bounded from both above and below. Hence, there must exist j and j' such that $J_+ |\lambda j\rangle = J_- |\lambda j'\rangle = 0 \Rightarrow N = 0$.

The above relations tells us $j(j+1) = j'(j'-1) = \lambda$, and thus $(j+j')(j-j'+1) = 0$. This gives two solutions: $j = -j'$ and $j = j' - 1$. However, as we define $j \geq j'$ via raising, so it is impossible that $j = j' - 1$. So, $j = -j'$.

As there are integer steps of raising, we say that

$$j = \frac{n}{2}, \quad (135)$$

which restricts the value of j to integers and half-integers.

From above discussions, we know $\lambda = j(j+1)$, so it is possible to relabel the eigenstates with j :

$$J^2 |jm\rangle = j(j+1) \hbar^2 |jm\rangle \quad (136)$$

$$J_3 |jm\rangle = m \hbar |jm\rangle, \quad m = -j, \dots, j \quad (2j+1 \text{ states}) \quad (137)$$

We may now rewrite the scale constants N :

$$\begin{aligned} |N_{\pm}|^2 &= [\lambda - m(m \pm 1)] \hbar^2 \\ &= [j(j+1) - m(m \pm 1)] \hbar^2 \\ &= [(j+m)(j-m) + j \mp m] \hbar^2 \end{aligned}$$

Therefore,

$$J_+ |jm\rangle = \hbar\sqrt{(j+m+1)(j-m)} |j(m+1)\rangle, \quad (138)$$

$$J_- |jm\rangle = \hbar\sqrt{(j-m+1)(j+m)} |j(m-1)\rangle. \quad (139)$$

7.5.3 Representations

Consider matrix elements $\langle m|\mathbf{J}|m'\rangle = \mathbf{J}_{mm'}$, where m stands for rows and m' for columns. Then, it is fairly straightforward to see:

$$\langle m'|J_3|m\rangle = m\hbar\delta_{mm'} \quad (140)$$

$$\langle m'|J_+|m\rangle = \hbar\sqrt{(j+m+1)(j-m)} \langle m'|m+1\rangle \quad (141)$$

$$\langle m'|J_-|m\rangle = \hbar\sqrt{(j-m+1)(j+m)} \langle m'|m-1\rangle \quad (142)$$

As linear combinations of J_+ and J_- , we know:

$$\langle m'|J_1|m\rangle = \frac{1}{2} (\langle m'|J_+|m\rangle + \langle m'|J_-|m\rangle) \quad (143)$$

$$\langle m'|J_2|m\rangle = \frac{1}{2i} (\langle m'|J_+|m\rangle - \langle m'|J_-|m\rangle) \quad (144)$$

- $j = 1/2$:

There are 2 eigenstates: $\left|\frac{1}{2}, \pm\frac{1}{2}\right\rangle$, or $|\uparrow\rangle, |\downarrow\rangle$.

By mapping

$$|\uparrow\rangle \mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle \mapsto \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (145)$$

$$\langle m'|J_3|m\rangle = m\hbar\delta_{mm'} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (146)$$

For J_+ , only $\langle \uparrow|J_+|\downarrow\rangle = (J_+)_{12} \neq 0$, $(J_+)_{12} = \hbar$. Similarly, only $(J_-)_{21} \neq 0$, $(J_-)_{21} = \hbar$. So, we get the representations of J_1 and J_2 :

$$\langle m'|J_1|m\rangle = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \langle m'|J_2|m\rangle = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (147)$$

At this point, we are able to conclude that

$$J_i = \frac{\hbar}{2} \sigma_i \quad (148)$$

in their 2D representations.

- $j = 1$:

There are 3 eigenstates in total, and mapping gives

$$|1, 1\rangle \mapsto \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |1, 0\rangle \mapsto \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |1, -1\rangle \mapsto \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Similar reasoning leads us to

$$\langle m' | J_3 | m \rangle = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad \langle m' | J_+ | m \rangle = \hbar \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix}. \quad (149)$$

Remember that $J_- = J_+^\dagger$, which may save half the work. Given the above,

$$\langle m' | J_- | m \rangle = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \langle m' | J_2 | m \rangle = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}. \quad (150)$$

In fact, these matrices are the only irreps of J with dimensions $2j + 1$. However, reps can be constructed via direct sum.

7.5.4 Integers & Half-integers

- Completeness Relation:

$$\mathbb{1} = \sum_{j=0} \sum_{m=-j}^j |jm\rangle \langle jm| \quad (151)$$

For a given j ,

$$\mathbb{1} = \sum_{m=-j}^j |jm\rangle \langle jm|. \quad (152)$$

- Effects of Rotation:

$$\mathcal{R}(\hat{\mathbf{z}}, 2\pi) |jm\rangle = e^{-i2\pi J_z/\hbar} |jm\rangle = e^{-i2\pi m} |jm\rangle = \begin{cases} |jm\rangle & \text{for integer } m (j) \text{ — bosons} \\ -|jm\rangle & \text{for half-integer } m (j) \text{ — fermions} \end{cases} \quad (153)$$

- $\mathcal{D}^j$ Matrices: reps of the SU group

$$\mathcal{D}_{m'm}^j [\mathcal{R}(\mathbf{n}, \omega)] \equiv \langle jm' | \mathcal{R}(\mathbf{n}, \omega) | jm \rangle \quad (154)$$

For $j = 1/2$,

$$\begin{aligned} \mathcal{D}_{m'm}^{1/2} [\mathcal{R}(\mathbf{n}, \omega)] &= \langle jm' | e^{-i\omega \mathbf{n} \cdot \mathbf{J}/\hbar} | jm \rangle \\ &= \langle jm' | e^{-i\omega \mathbf{n} \cdot \boldsymbol{\sigma}/2} | jm \rangle \\ &= \langle jm' | \cos\left(-\frac{\omega}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right) + i \sin\left(-\frac{\omega}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right) | jm \rangle. \end{aligned}$$

Now,

$$\begin{aligned} (\mathbf{n} \cdot \boldsymbol{\sigma})^2 &= n_i n_j \sigma_i \sigma_j \\ &= n_i n_j (\delta_{ij} + i \varepsilon_{ijk} \sigma_k) \\ &= n_i n_i + i \mathbf{n} \cdot (\mathbf{n} \times \boldsymbol{\sigma}) = 1, \end{aligned}$$

$$\mathcal{D}_{m'm}^{1/2} [\mathcal{R}(\mathbf{n}, \omega)] = \cos\left(\frac{\omega}{2}\right) \mathbb{1} - i \sin\left(\frac{\omega}{2}\right) \mathbf{n} \cdot \boldsymbol{\sigma}. \quad (155)$$

This gives $\mathcal{D}_{m'm}^{1/2} [\mathcal{R}(\mathbf{n}, 2\pi)] = -\mathbb{1}$ as expected.

There are two properties of these special $\mathcal{D}$ matrices. Firstly, they are unitary, as suggested by themselves being the elements of SU group. Then,

$$\sum_{m''} \mathcal{D}_{m'm''}^j (\mathcal{R}_2) \mathcal{D}_{m''m}^j (\mathcal{R}_1) = \mathcal{D}_{m'm}^j (\mathcal{R}_2 \mathcal{R}_1). \quad (156)$$

Proof. By completeness relation given j ,

$$\mathcal{D}_{m'm}^j (\mathcal{R}_2 \mathcal{R}_1) = \langle jm' | \mathcal{R}_2 \mathcal{R}_1 | jm \rangle = \sum_{m''} \langle jm' | \mathcal{R}_2 | jm'' \rangle \langle jm'' | \mathcal{R}_1 | jm \rangle.$$

□

8 Tensor Products & Addition of Angular Momentum

8.1 Tensor Products

8.1.1 Definition

Tensor products construct new Hilbert spaces from old ones.

For N -dimensional $\mathcal{H}^{(1)} \{|n\rangle\}$ and M -dimensional $\mathcal{H}^{(2)} \{|\alpha\rangle\}$,

$$\mathcal{H} \equiv \mathcal{H}^{(1)} \otimes \mathcal{H}^{(2)} \quad (157)$$

has $N \times M$ dimensions with basis

$$|n\rangle \otimes |\alpha\rangle = |n\rangle |\alpha\rangle = |n, \alpha\rangle. \quad (158)$$

8.1.2 Inner Product

$$(|n\rangle |\alpha\rangle)^\dagger \equiv \langle \alpha | \langle n | \quad (159)$$

With this in mind, we can easily see that

$$(\langle \alpha | \langle n |) (|m\rangle |\beta\rangle) = \langle n | m \rangle \langle \alpha | \beta \rangle. \quad (160)$$

8.1.3 Operators

Firstly, consider

$$A \equiv A^{(1)} \otimes \mathbb{1},$$

where $A : \mathcal{H} \rightarrow \mathcal{H}$, $A^{(1)} : \mathcal{H}^{(1)} \rightarrow \mathcal{H}^{(1)}$:

$$A (|n\rangle |\alpha\rangle) \equiv \left(A^{(1)} |n\rangle \right) |\alpha\rangle.$$

Similarly, one may define $B \equiv \mathbb{1} \otimes B^{(2)}$. Then,

$$\begin{aligned} AB |x\rangle |\gamma\rangle &= \left(A^{(1)} \otimes \mathbb{1} \right) \left(\mathbb{1} \otimes B^{(2)} \right) |x\rangle |\gamma\rangle \\ &= \left(A^{(1)} \otimes \mathbb{1} \right) |x\rangle \left(B^{(2)} |\gamma\rangle \right) \\ &= \left(A^{(1)} |x\rangle \right) \left(B^{(2)} |\gamma\rangle \right). \end{aligned}$$

Clearly, $[A, B] = 0$.

More generally, we may define $A^{(1)} \otimes B^{(2)} : \mathcal{H} \rightarrow \mathcal{H}$:

$$A^{(1)} \otimes B^{(2)} (|n\rangle |\alpha\rangle) \equiv \left(A^{(1)} |n\rangle \right) \left(B^{(2)} |\alpha\rangle \right). \quad (161)$$

8.2 Addition of Angular Momentum

In this section, we assume $\hbar = 1$. Consider the tensor product of two spin-1/2 Hilbert spaces. Given $\mathcal{H}^{(1)} : \{|\uparrow\rangle_1, |\downarrow\rangle_1\}$, $\mathcal{H}^{(2)} : \{|\uparrow\rangle_2, |\downarrow\rangle_2\}$,

$$\mathcal{H} : \{|\uparrow\rangle_1 |\uparrow\rangle_2, |\uparrow\rangle_1 |\downarrow\rangle_2, |\downarrow\rangle_1 |\uparrow\rangle_2, |\downarrow\rangle_1 |\downarrow\rangle_2\}.$$

Then, we define the total spin operator:

$$\mathbf{S} = \mathbf{S}^{(1)} + \mathbf{S}^{(2)} \left(= \mathbf{S}^{(1)} \otimes \mathbf{1} + \mathbf{1} \otimes \mathbf{S}^{(2)} \right). \quad (162)$$

It is trivial to check that this new spin operator still satisfies the commutation relation:

$$\begin{aligned} [S_i, S_j] &= [S_i^{(1)} + S_i^{(2)}, S_j^{(1)} + S_j^{(2)}] \\ &= [S_i^{(1)}, S_j^{(1)}] + [S_i^{(2)}, S_j^{(2)}] \\ &= i\varepsilon_{ijk} S_k^{(1)} + i\varepsilon_{ijk} S_k^{(2)} \\ &= i\varepsilon_{ijk} S_k. \end{aligned}$$

8.2.1 Orthonormal Basis for Total Spin

We want $|sm\rangle$ for S^2 and S_3 . From our previous labeling,

$$S_3 |m_1\rangle_1 |m_2\rangle_2 = \left(S_3^{(1)} |m_1\rangle_1 \right) \left(S_3^{(2)} |m_2\rangle_2 \right) = (m_1 + m_2) |m_1\rangle_1 |m_2\rangle_2,$$

so $|m_1\rangle_1 |m_2\rangle_2$ is already the eigenstate of S_3 .

	$ \uparrow\rangle_1 \uparrow\rangle_2$	$ \uparrow\rangle_1 \downarrow\rangle_2$	$ \downarrow\rangle_1 \uparrow\rangle_2$	$ \downarrow\rangle_1 \downarrow\rangle_2$
$m = m_1 + m_2$	1	0	0	-1

The eigenvalues thus suggest that the quantum number $s = 1$ or 1. Then, it is trivial to see

$$|s = 1, m = 1\rangle = |\uparrow\rangle_1 |\uparrow\rangle_2, \quad |s = 1, m = -1\rangle = |\downarrow\rangle_1 |\downarrow\rangle_2. \quad (163)$$

To find $|s = 1, m = 0\rangle$, remember that this state may be derived via raising and lowering:

$$\begin{aligned} S_+ |s = 1, m = -1\rangle &= \sqrt{2} |s = 1, m = 0\rangle \\ &= (S_+^{(1)} + S_+^{(2)}) |\downarrow\rangle_1 |\downarrow\rangle_2 \\ &= |\uparrow\rangle_1 |\downarrow\rangle_2 + |\downarrow\rangle_1 |\uparrow\rangle_2. \end{aligned}$$

Therefore, we find that the real $|s = 1, m = 0\rangle$ is actually the mixed state of $m = 0$ in the table:

$$|s = 1, m = 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_1 |\downarrow\rangle_2 + |\downarrow\rangle_1 |\uparrow\rangle_2). \quad (164)$$

For $|s = 0, m = 0\rangle$, $m = 0$ means that this state should also be the linear combination of $|\uparrow\rangle_1 |\downarrow\rangle_2$ and $|\downarrow\rangle_1 |\uparrow\rangle_2$. Additionally, with different s , $\langle s = 1, m = 0 | s = 0, m = 0 \rangle = 0$. Thus,

$$|s = 0, m = 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_1 |\downarrow\rangle_2 - |\downarrow\rangle_1 |\uparrow\rangle_2) \quad (165)$$

up to a phase.

With the above derivations, we see that the new basis is actually $\mathcal{H}^{s=1} \oplus \mathcal{H}^{s=0}$, where $\oplus$ denotes the direct sum:

$$\mathcal{H}^{s=1/2} \otimes \mathcal{H}^{s=1/2} = \mathcal{H}^{s=1} \oplus \mathcal{H}^{s=0}. \quad (166)$$

When viewing in dimensions, we see

$$2 \times 2 = 3 + 1.$$

8.2.2 Tensor Product & Direct Sum

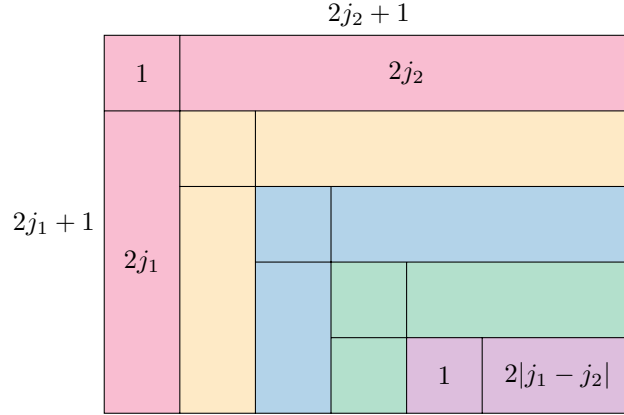
If we are to add three spins together, note that the operation of tensor product and direct sum is distributive:

$$\mathcal{H}^{s=1/2} \otimes \mathcal{H}^{s=1/2} \otimes \mathcal{H}^{s=1/2} = \mathcal{H}^{s=1/2} \otimes (\mathcal{H}^{s=1} \oplus \mathcal{H}^{s=0}) = (\mathcal{H}^{s=3/2} \oplus \mathcal{H}^{s=1/2}) \oplus \mathcal{H}^{s=1/2}.$$

In general,

$$\mathcal{H}^{j_1} \otimes \mathcal{H}^{j_2} = \mathcal{H}^{j_1+j_2} \oplus \mathcal{H}^{j_1+j_2-1} \oplus \dots \oplus \mathcal{H}^{|j_1-j_2|}. \quad (167)$$

The result in dimensions is much easier to comprehend when viewing in a simple visualization.



From this, it is easy to see that

$$\begin{aligned}
 (2j_1 + 1)(2j_2 + 1) &= (2j_1 + 2j_2 + 1) + (2j_1 + 2j_2 + 1 - 2) + \cdots + (2|j_1 - j_2| + 1) \\
 &= [2(j_1 + j_2) + 1] + [2(j_1 + j_2 - 1) + 1] + [2(j_1 + j_2 - 2) + 1] + \cdots + (2|j_1 - j_2| + 1).
 \end{aligned}$$

8.2.3 Clebsch-Gordon Coefficients

Clebsch-Gordon coefficients are the relation between two orthonormal bases:

$$\begin{cases} |j_1, j_2; m_1, m_2\rangle \equiv |j_1, m_1\rangle \otimes |j_2, m_2\rangle : & (J^{(1)})^2, (J^{(2)})^2, J_3^{(1)}, J_3^{(2)} \\ |j_1, j_2; j, m\rangle : & (J^{(1)})^2, (J^{(2)})^2, J^2, J_3 \end{cases} \quad (168)$$

By completeness relation,

$$\begin{aligned}
 |j_1, j_2; j, m\rangle &= \left(\sum_{m_1=-j_1}^{j_1} |j_1 m_1\rangle \langle j_1 m_1| \right) \otimes \left(\sum_{m_2=-j_2}^{j_2} |j_2 m_2\rangle \langle j_2 m_2| \right) |j_1, j_2; j, m\rangle \\
 &= \sum_{m_1 m_2} |j_1, j_2; j, m\rangle \langle j_1, j_2; m_1, m_2 | j_1, j_2; j, m \rangle,
 \end{aligned} \quad (169)$$

where the coefficients in orange are called Clebsch-Gordon coefficients.

8.2.4 Single Particle with Spin

The total angular momentum is defined as the sum of orbital angular momentum and spin angular momentum:

$$\mathbf{J} = \mathbf{L} + \mathbf{S}. \quad (170)$$

One choice of basis is $|\mathbf{x}\rangle \otimes |sm\rangle = |\mathbf{x}, m\rangle$, where s is fixed and, thus, abbreviated.

The wavefunction, $\psi(\mathbf{x}, m) \equiv \langle \mathbf{x}, m | \psi \rangle$, may be represented as a column vector. The rotation operator is

$$\begin{aligned}\mathcal{R}(\mathbf{n}, \omega) |\psi\rangle &= e^{-i\omega \mathbf{n} \cdot \mathbf{J} / \hbar} |\psi\rangle \\ &= e^{-i\omega \mathbf{n} \cdot \mathbf{L} / \hbar} e^{-i\omega \mathbf{n} \cdot \mathbf{S} / \hbar} |\psi\rangle,\end{aligned}\tag{171}$$

as $[L, S] = 0$.

Then, the effect of rotation is given by:

$$\begin{aligned}\langle \mathbf{x}, m | \mathcal{R}(\mathbf{n}, \omega) | \psi \rangle &= \langle \mathbf{x}, m | e^{-i\omega \mathbf{n} \cdot \mathbf{L} / \hbar} e^{-i\omega \mathbf{n} \cdot \mathbf{S} / \hbar} | \psi \rangle \\ &= \langle \mathbf{x} | e^{-i\omega \mathbf{n} \cdot \mathbf{L} / \hbar} \langle m | e^{-i\omega \mathbf{n} \cdot \mathbf{S} / \hbar} | \psi \rangle \\ &= \langle R^{-1} \mathbf{x} | \sum_{m'} \langle m | e^{-i\omega \mathbf{n} \cdot \mathbf{S} / \hbar} | m' \rangle \langle m' | \psi \rangle \\ &= \langle R^{-1} \mathbf{x} | \sum_{m'} \mathcal{D}_{mm'}^s(\mathcal{R}) \langle m' | \psi \rangle \\ &= \sum_{m'} \mathcal{D}_{mm'}^s(\mathcal{R}) \psi(R^{-1} \mathbf{x}, m').\end{aligned}$$

If we denote $|\psi^{\mathcal{R}}\rangle \equiv \mathcal{R}|\psi\rangle$, then

$$\psi_m^{\mathcal{R}}(R\mathbf{x}) = \sum_{m'} \mathcal{D}_{mm'}^s(\mathcal{R}) \psi_{m'}(\mathbf{x}).$$

9 Density Operators and Entanglement

9.1 Density Operators

9.1.1 Definition

A density operator ρ is a linear operator. In general,

$$\rho = \sum_i p_i |\phi_i\rangle\langle\phi_i| \quad (172)$$

with $\sum_i p_i = 1$, $p_j \geq 0$. There are several important properties for a density operator.

- $\rho^\dagger = \rho$;
- $\text{Tr } \rho = 1$, which is actually $\sum_i p_i = 1$;

$$\begin{aligned} \text{Tr } \rho &= \sum_n \langle n | \rho | n \rangle = \sum_{in} p_i \langle n | \phi_i \rangle \langle \phi_i | n \rangle \\ &= \sum_{in} p_i \langle \phi_i | n \rangle \langle n | \phi_i \rangle \\ &= \sum_i p_i \langle \phi_i | \phi_i \rangle = \sum_i p_i = 1. \end{aligned}$$

- $\langle \psi | \rho | \psi \rangle \geq 0$, $\forall |\psi\rangle$, which is $p_j \geq 0$;

$$\langle \psi | \rho | \psi \rangle = \sum_i p_i \langle \psi | \phi_i \rangle \langle \phi_i | \psi \rangle = \sum_i p_i |\langle \phi_i | \psi \rangle|^2 \geq 0.$$

- $\text{Tr } \rho^2 \leq 1$ with equality iff ρ describes a pure state;

$$\begin{aligned} \text{Tr } \rho^2 &= \sum_n \langle n | \left(\sum_{ij} p_i p_j |\phi_i\rangle\langle\phi_i| |\phi_j\rangle\langle\phi_j| \right) | n \rangle \\ &= \sum_{ijmn} p_i p_j \langle n | \phi_i \rangle \langle \phi_i | m \rangle \langle m | \phi_j \rangle \langle \phi_j | n \rangle \\ &= \sum_{ij} p_i p_j |\langle \phi_i | \phi_j \rangle|^2 \\ &\begin{cases} = \sum_i p_i^2 \leq \sum_i p_i = 1 \\ \leq \sum_{ij} p_i p_j \langle \phi_i | \phi_i \rangle \langle \phi_j | \phi_j \rangle = \sum_i p_i \sum_j p_j = 1 \end{cases} \end{aligned}$$

The Schwarz inequality takes the equal sign when $|\phi_i\rangle = \alpha |\phi_j\rangle$. So, $\langle\phi_i|\phi_i\rangle = \langle\phi_j|\phi_j\rangle = 1$ implies that $|\alpha|^2 = 1$. Therefore,

$$\rho = |\psi\rangle\langle\psi| \quad (173)$$

for a pure state.

9.1.2 Postulates of QM in the Language of Density Operators

- Every state is associated with a density operator;
- $P(A = a_n) = \text{Tr}(\rho P_n)$;

Proof. In the most general case,

$$\begin{aligned} \text{Tr}(\rho P_n) &= \sum_{m, r_m} \langle m, r_m | \left(\sum_i p_i |\phi_i\rangle\langle\phi_i| \sum_{r_n} |n, r_n\rangle\langle n, r_n| \right) | m, r_m \rangle \\ &= \sum_{i, m, r_m, r_n} p_i \langle m, r_m | \phi_i \rangle \langle \phi_i | n, r_n \rangle \langle n, r_n | m, r_m \rangle \\ &= \sum_{i, m, r_m, r_n} p_i \langle m, r_m | \phi_i \rangle \langle \phi_i | n, r_n \rangle \delta_{mn} \delta_{r_m r_n} \\ &= \sum_{i, r_n} p_i |\langle n, r_n | \phi_i \rangle|^2 = P(A = a_n) \end{aligned}$$

□

- $\langle A \rangle = \text{Tr}(\rho A)$

Proof. By the spectral theorem, $A = \sum_n a_n P_n$,

$$\text{tr}(\rho A) = \sum_n a_n \text{Tr}(\rho P_n) = \sum_n a_n P(A = a_n) = \langle A \rangle.$$

□

- After a measurement with outcome a_n , the system collapses into the state

$$\rho \rightarrow \frac{P_n \rho P_n}{\text{Tr}(\rho P_n)}. \quad (174)$$

It helps to consider measuring $A = a_n$ again after the initial measurement:

$$\begin{aligned} P(A = a_n) &= \frac{1}{\text{Tr}(\rho P_n)} \text{Tr}(P_n \rho P_n \cdot P_n) \\ &= \frac{1}{\text{Tr}(\rho P_n)} \text{Tr}(P_n \rho P_n) \\ &= \frac{1}{\text{Tr}(\rho P_n)} \text{Tr}(\rho P_n) = 1. \end{aligned}$$

- Time Evolution:

Before giving the formula, we should think of the Schrödinger picture, which tells us that

$$|\psi, t\rangle = U(t, t_0) |\psi, t_0\rangle. \quad (175)$$

Therefore, naturally we know that the density operator evolves by

$$\rho(t) = e^{-iHt/\hbar} \rho(0) e^{iHt/\hbar}, \quad (176)$$

given $\partial_t H = 0$. This gives us the equation of time evolution:

$$i\hbar \frac{d}{dt} \rho(t) = -[\rho, H]. \quad (177)$$

Remember that this is different from Heisenberg equation of motion (by a minus sign). This makes sense, as all our conclusions by now are based on the Schrödinger picture. The density operators are not technically operators.

In the Heisenberg picture, the states do not evolve, neither does the density operator.

9.2 Thermal Systems

$$\rho = \frac{1}{Z} e^{-\beta H} \quad (178)$$

To satisfy $\text{Tr} \rho = 1$, we have

$$\begin{aligned} Z &= \text{Tr}(e^{-\beta H}) \\ &= \sum_{nr_n} \langle n, r_n | e^{-\beta H} | n, r_n \rangle \\ &= \sum_{nr_n} e^{-\beta E_n}, \end{aligned}$$

$$Z = \sum_n d_n e^{-\beta E_n}. \quad (179)$$

9.3 Reduced Density Operators

Consider $\rho : \mathcal{H} \rightarrow \mathcal{H}$ where $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$.

The reduced density operator ρ_A for system A is obtained by tracing over the states in the Hilbert space for B :

$$\rho_A \equiv \text{Tr}_{\mathcal{H}_B} \rho : \mathcal{H}_A \rightarrow \mathcal{H}_A. \quad (180)$$

Now,

$$\begin{aligned} \text{Tr } C &= \sum_{\alpha n} \langle \alpha | \langle n | C | n \rangle | \alpha \rangle \\ &= \sum_{\alpha} \langle \alpha | \text{Tr}_{\mathcal{H}_B} C | \alpha \rangle \\ &= \text{Tr}_{\mathcal{H}_A} (\text{Tr}_{\mathcal{H}_B} C). \end{aligned} \quad (181)$$

Thus, for $C = C_A \otimes \mathbf{1}$,

$$\begin{aligned} \langle C \rangle &= \text{Tr} (\rho C) \\ &= \text{Tr}_{\mathcal{H}_A} [\text{Tr}_{\mathcal{H}_B} (\rho (C_A \otimes \mathbf{1}))] \\ &= \text{Tr}_{\mathcal{H}_A} [(\text{Tr}_{\mathcal{H}_B} \rho) C_A] \\ &= \text{Tr}_{\mathcal{H}_A} (\rho_A C_A). \end{aligned} \quad (182)$$

9.4 Entanglement

While ρ may satisfy $\text{Tr } \rho^2 = 1$, ρ_A may not.

If $\text{Tr } \rho^2 = 1$, but $\text{Tr } \rho_A^2 \neq 1$, then we say that ρ_A is entangled.

Consider the tensor product of 2 qubits.

- Factorizable states are not entangled.

We say a state $|\psi\rangle$ is factorizable if $|\psi\rangle = |\psi\rangle_A \otimes |\psi\rangle_B$. As an example, we know $|0, 0\rangle = |0\rangle \otimes |0\rangle$.

Therefore,

$$\begin{aligned} \rho_A &= \text{Tr}_{\mathcal{H}_B} \rho \\ &= \text{Tr}_{\mathcal{H}_B} (|0\rangle_A \langle 0|_A \otimes |0\rangle_B \langle 0|_B) \\ &= |0\rangle_A \langle 0|_A, \end{aligned}$$

which is a non-entangled pure state.

- Bell States: One of the Bell states is $|\psi\rangle = \frac{1}{\sqrt{2}}(|0,0\rangle + |1,1\rangle)$. Then,

$$\begin{aligned}
\rho_A &= \text{Tr}_{\mathcal{H}_B} \rho \\
&= \frac{1}{2} \langle 0|_B (|0,0\rangle + |1,1\rangle) (\langle 0,0| + \langle 1,1|) |0\rangle_B \\
&\quad + \frac{1}{2} \langle 1|_B (|0,0\rangle + |1,1\rangle) (\langle 0,0| + \langle 1,1|) |1\rangle_B \\
&= \frac{1}{2} (|0\rangle_A \langle 0|_A + |1\rangle_A \langle 1|_A).
\end{aligned}$$

As we see that

$$\rho_A^2 = \frac{1}{4} (|0\rangle_A \langle 0|_A + |1\rangle_A \langle 1|_A)^2 = \frac{1}{2} \rho_A,$$

$\text{Tr} \rho_A^2 = 1/2$, which means that ρ_A is entangled.

Other Bell states for 2 qubits are

$$\frac{1}{\sqrt{2}} (|0,0\rangle - |1,1\rangle), \quad \frac{1}{\sqrt{2}} (|0,1\rangle + |1,0\rangle), \quad \frac{1}{\sqrt{2}} (|0,1\rangle - |1,0\rangle). \quad (183)$$

9.5 Entanglement Entropy

9.5.1 Classical Shannon Entropy

$$S = - \sum_i p_i \ln p_i \quad \left(= - \frac{1}{k_B} \text{Gibbs Entropy} \right) \quad (184)$$

9.5.2 Quantum von-Neumann Entropy

$$S = - \text{Tr} (\rho \log \rho) \quad (185)$$

If ρ is the reduced density operator for a bipartite system, then S is called the **entanglement entropy**.

Given $\rho = \sum_i p_i |i\rangle\langle i|$,

$$\rho \log \rho = \sum_i p_i \ln p_i |i\rangle\langle i|. \quad (186)$$

Thus,

$$\begin{aligned}
S &= - \text{Tr} (\rho \log \rho) \\
&= - \sum_i p_i \ln p_i \sum_j \langle j|i\rangle \langle i|j\rangle \\
&= - \sum_i p_i \ln p_i,
\end{aligned}$$

which returns to the classical formula.

9.5.3 Properties of von-Neumann Entropy

- $S(\rho) \geq 0$;
- $S = 0 \Leftrightarrow \rho$ is a pure state;
- $\max(S) = \ln N$ when $\rho = \frac{1}{N} \mathbb{1} = \frac{1}{N} \sum_i |i\rangle\langle i|$;

- S is concave:

$$S\left(\sum_i \lambda_i \rho_i\right) \geq \sum_i \lambda_i S(\rho_i) \quad \text{with} \quad \sum_i \lambda_i = 1, \lambda_j \geq 0 \quad (187)$$

- Sub-additivity:

$$S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B) \quad (188)$$

The entropy of the whole is less than the sum of the parts.

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}) \quad (189)$$

9.6 Tensor Product of Simple Harmonic Oscillators

Consider $b_1 = a \otimes \mathbb{1}$, $b_2 = \mathbb{1} \otimes a$, and the pure state $|\psi\rangle = ce^{\alpha b_1^\dagger b_2^\dagger} |0, 0\rangle$. Calculating the raising operators,

$$\begin{aligned} |\psi\rangle &= ce^{\alpha b_1^\dagger b_2^\dagger} |0, 0\rangle \\ &= c \sum_n \frac{\alpha^n}{n!} (b_1^\dagger b_2^\dagger)^n |0, 0\rangle \\ &= c \sum_n \frac{\alpha^n}{n!} (\sqrt{n!})^2 |n, n\rangle \\ &= c \sum_n \alpha^n |n, n\rangle. \end{aligned}$$

Then,

$$\begin{aligned} \langle\psi|\psi\rangle &= |c|^2 \left(\sum_m \alpha^{*m} \langle m, m| \right) \left(\sum_n \alpha^n |n, n\rangle \right) \\ &= |c|^2 \sum_n |\alpha|^{2n} \\ &= \frac{|c|^2}{1 - |\alpha|^2} \quad (|\alpha| < 1). \end{aligned}$$

Therefore, we may choose $|c|^2 = 1 - |\alpha|^2$ such that $\langle\psi|\psi\rangle = 1$.

The density operator is

$$\rho = |\psi\rangle\langle\psi| = |c|^2 \sum_{mn} \alpha^n \alpha^{*m} |n, n\rangle\langle m, m|. \quad (190)$$

The reduced density operator is

$$\begin{aligned} \rho_1 &= \text{Tr}_{\mathcal{H}_2} \rho \\ &= |c|^2 \sum_{mnk} \alpha^n \alpha^{*m} |n\rangle_1 \langle m|_1 (\langle k|n\rangle_2 \langle m|k\rangle_2) \\ &= |c|^2 \sum_{mnk} \alpha^n \alpha^{*m} |n\rangle_1 \langle m|_1 \delta_{nk} \delta_{mk} \\ &= |c|^2 \sum_n |\alpha|^2 |n\rangle_1 \langle n|_1 \\ &= (1 - |\alpha|^2) \sum_n e^{-n \ln(1/|\alpha|^2)} |n\rangle_1 \langle n|_1. \end{aligned}$$

By defining $\ln \frac{1}{|\alpha|^2} = \hbar\omega\beta$,

$$\rho_1 = (1 - e^{-\hbar\omega\beta}) \sum_n e^{-n\hbar\omega\beta} |n\rangle\langle n| = (e^{\hbar\omega\beta/2} - e^{-\hbar\omega\beta/2}) \sum_n e^{-\beta E_n} |n\rangle\langle n|, \quad (191)$$

where $E_n = \hbar\omega \left(n + \frac{1}{2}\right)$.

Now, note that

$$\begin{aligned} \text{Tr}(e^{-\beta H_1}) &= \sum_n \langle n|e^{-\beta H_1}|n\rangle \\ &= \sum_n e^{-\beta E_n} \\ &= e^{-\hbar\omega\beta/2} \sum_n e^{-n\hbar\omega\beta} \\ &= e^{-\hbar\omega\beta/2} \frac{1}{1 - e^{-\hbar\omega\beta}} \\ &= (e^{\hbar\omega\beta/2} - e^{-\hbar\omega\beta/2})^{-1}, \end{aligned}$$

so we see

$$\rho_1 = \frac{e^{-\beta H_1}}{\text{Tr}(e^{-\beta H_1})}. \quad (192)$$

As an reminder here, the state $|\psi\rangle$ is

$$|\psi\rangle = (1 - e^{-\hbar\omega\beta})^{1/2} e^{-b_1^\dagger b_2^\dagger \hbar\omega\beta/2} |0, 0\rangle. \quad (193)$$

One may check that the reduced density operator ρ_1 is entangled and purely thermal.

9.7 Functions of a Matrix

For a Hermitian diagonalizable matrix ρ ,

$$f(\rho) = U f(D) U^\dagger, \quad (194)$$

with $\rho = U D U^\dagger$ where D is diagonal. $f(D)$ is acting f on each eigenvalue.

The spirit is that **analytic functions of matrices are defined only via Taylor expansion**, but not directly. The common functions we know from Taylor expansion are $\log \rho$, $\exp \rho$, $\sqrt{\rho}$, ρ^{-1} (except positive integer powers).

For positive integer powers, it is not necessary to diagonalize:

$$\text{Tr } \rho = \text{Tr } (U D U^\dagger) = \text{Tr } (D U^\dagger U) = \text{Tr } D, \quad (195)$$

$$\text{Tr } \rho^2 = \text{Tr } (U D U^\dagger U D U^\dagger) = \text{Tr } D^2. \quad (196)$$

It seems that all functions expandable as Taylor series do not need diagonalization. However, one cannot guarantee if the expansion is **convergent**. Therefore, the easiest way is via diagonalization.

Also, measuring probability without prior knowledge of the eigenstates needs diagonalization.

$$P(\lambda_i) = \text{Tr}(\rho |\psi_i\rangle\langle\psi_i|) = \langle\psi_i|\rho|\psi_i\rangle \quad (197)$$

The above equation seems that there is no need for diagonalization; however, without knowing $|\psi_i\rangle$, we need to diagonalize ρ to find eigenstates.

As an example,

$$\rho = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Diagonalization yields

$$\begin{aligned} \lambda_1 = 1: \quad |\psi_1\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \\ \lambda_2 = 3: \quad |\psi_2\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \end{aligned}$$

From this, we can also reconstruct the density operator:

$$\rho = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|. \quad (198)$$

Remember that we need $\text{Tr } \rho = 1$ — this means that the eigenvalues λ_i are exactly the probabilities for the diagonalized states.

In order to calculate the entropy, diagonalization is needed for reasons like convergence. Consider the density matrix

$$\rho = \frac{1}{6} \begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix}.$$

The eigenvalues are $3 \pm \sqrt{5}$, which means that

$$\begin{aligned} S &= -\text{Tr}(\rho \log \rho) \\ &= -\sum_i p_i \ln p_i \\ &= -\left(\frac{1}{2} + \frac{1}{6}\sqrt{5}\right) \ln \left(\frac{1}{2} + \frac{1}{6}\sqrt{5}\right) - \left(\frac{1}{2} - \frac{1}{6}\sqrt{5}\right) \ln \left(\frac{1}{2} - \frac{1}{6}\sqrt{5}\right). \end{aligned}$$

Given this, we see that knowing p_i without really diagonalizing the states are already enough for calculating the entropy.

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